

THE OPERATIONAL MANUAL OF ELECTRIC CHAIN HOIST



Carefully keep the operational manual and as reference

Safety First

Operating hoisting equipment can potentially cause damage to property or personal injury. Operators must follow the proper operating instructions and pay attention to the precautions in this manual to reduce this risk and ensure the safe operation of the equipment.



This warning sign indicates a potentially high risk operation or critical safety instruction. Operators must fully comply with the instructions and precautions in these sections.

Please read through the entire manual prior to using this electric hoist.

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1. Introduction

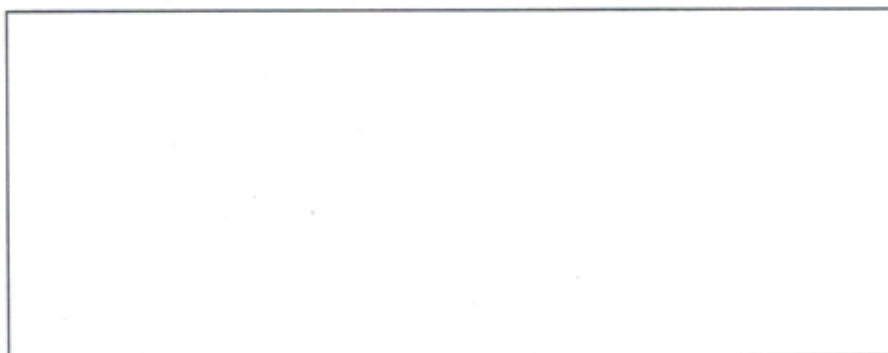
This manual will assist the operator to correctly install, operate and maintain the electric chain hoist.

Lifting equipment and machines must only be operated by competent personnel and only once this manual has been read.

In order to provide the replacement parts in the shortest time, please provide the following information:

- a. The type and model of the hoist
- b. The serial number of the hoist
- c. The name and code of the part as detailed in this manual

If you have any questions please don't hesitate to contact us.



(Distributor details)

2. Hoist specifications

2.1 Table of specifications

The specifications in the following table are applicable to all electric chain hoists.

Table 2.1 Specifications

Item		Specification	
Range of working humidity		85% or below 85%	
Range of working temperature		-20°C-to 40°C	
Protection class	Hoist	Ip54	
	Push button	Ip54	
Power supply		3 phase,200-600V,50/60Hz	
Noise degree	Single speed hoist	81dB	
	Double speed hoist	81dB	
Chain specifications		Nominal dia	Internal length
	0.3t,0.5t	6.3mm	19mm
	1t,2t,3t	7.1mm	21mm
	1.5t,2t,3t	10.0mm	30mm
	2.5t,3t,5t,7.5t	11.2mm	34mm

Remarks

If the hoist is working in an environment where the temperature and humidity are outside the range in the table, please contact the distributor for related information.

This electric chain hoist is designed to lift vertically in a normal atmosphere and under normal working conditions.

The noise level was measured at a distance of 1 meter from the hoist under normal working conditions.

2.2 Machine usage and service life

The safety and service life of the chain hoist can only be guaranteed if the unit is operated according to the following:

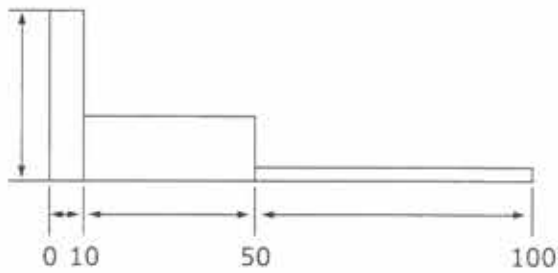
The design of the hoist conforms to 1Am of FEC (table 2.21)

The load condition is defined in table 2.2

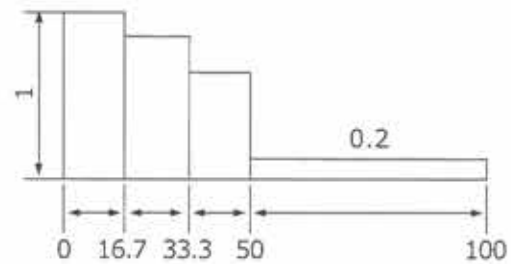
The average operating hours per day and total operating hours are calculated on the load distribution.

Table 2.2 Categories of machine usage

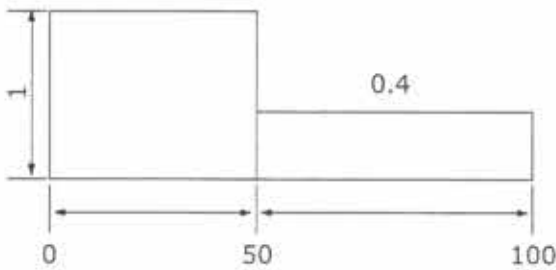
Load types	Definition	Cubic meters	Average operating hours / day(hours)					
1 (light duty)	The hoist and parts are frequently under light load and no max. load is lifted other than under exceptional conditions	$K \leq 0.50$	≤ 2	2-4	4-8	8-16	≤ 16	> 16
2 (medium duty)	The hoist and parts are frequently under light load and no max. load is lifted other than under exceptional conditions	$0.50 < K \leq 0.63$	≤ 1	1-2	2-4	4-8	8-16	≤ 16
3 (heavy duty)	The hoist and parts are frequently under medium and heavy load	$0.63 < K \leq 0.80$	≤ 0.5	0.5-1	1-2	2-4	4-8	8-16
4 (very heavy duty)	The hoist and parts are frequently under max. or near max. load	$0.80 < K \leq 1.00$	≤ 0.25	0.25-0.5	0.5-1	1-2	2-4	4-8
			1Bm	1Am	2m	3m	4m	5m



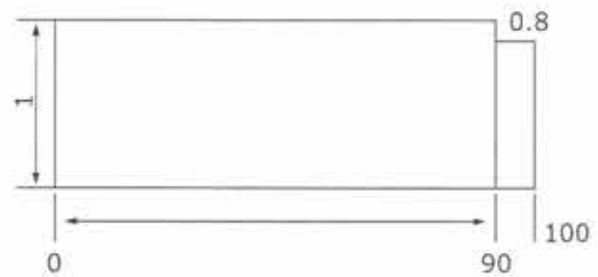
Operating time-load condition 1



Operating time-load condition 2



Operating time-load condition 3



Operating time-load condition 4

Table 2.21 Standard of lifting motor

Group		Intermittent operation			Continuous Operation
F.E.M	ISO	Cycles/h	Starts/h	ED(%)	Operation cycle per min
1 DM	M1	15	90	15	7.5
1 CM	M2	20	120	20	7.5
1 BM	M3	25	250	25	15
1 AM	M4	30	180	30	15
2 M	M5	40	240	40	30
3 M	M6	50	300	50	30
4 M	M7	60	360	60	60
5 M	M8	60	360	60	>60

2.3 Safety devices

Side magnetic braking device

The unique design causes the brake to activate immediately should the power be cut to the hoist.

Hooks and safety latches

The hook is hot forged using high tensile steel and heat treated to meet the demands on strength and hardness. The 360 degrees rotation and safety of the lower hook ensures operator safety.

Anti-phase protecting device

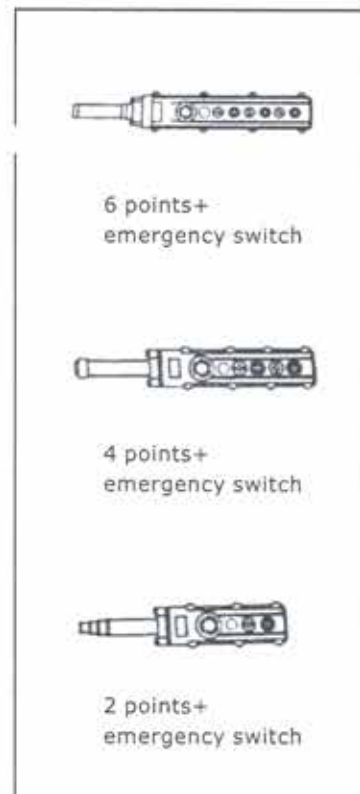
The design of the circuit protects the hoist by ensuring the electric motor doesn't burn out should the hoist be wired incorrectly to the power supply.

Limit switch

This switch is fitted to stop the motor automatically should the load chain extend beyond the safe length.

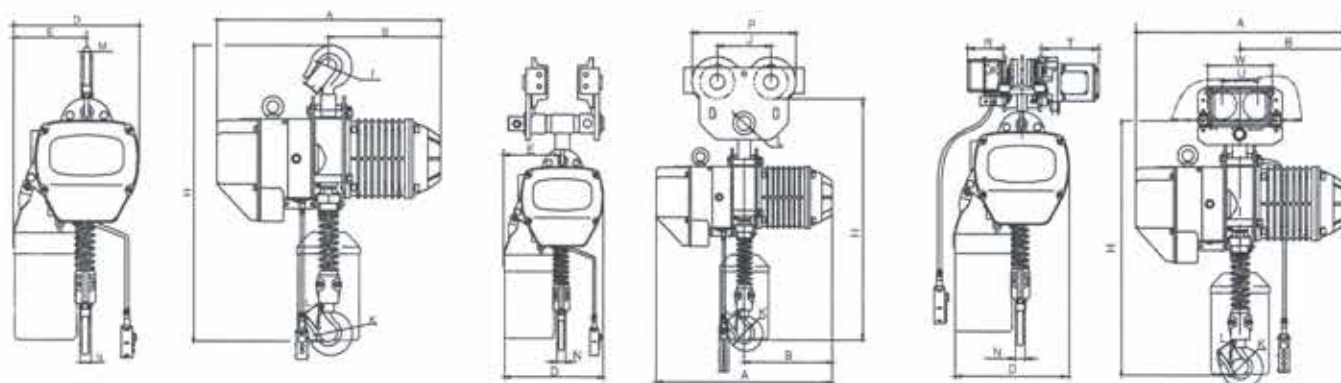
Emergency stop switch (optional)

This button is used to stop the hoist in an emergency. It is a red, mushroom shaped push button located at the top of the control pendant. When the button is pushed the power supply to the hoist is cut off and the button is automatically locked. Turn the button clockwise to release the button and restart the hoist.



2.4 Technical parameters-electric chain hoist

Capacity(ton)	0.3	0.5	1	1	1.5	2	2	2.5	3	3	3	5	7.5
Model	PV-HKD 00301S	PV-HKD 00501S	PV-HKD 0102S	PV-HKD 0101S	PV-HKD 01501S	PV-HKD 0201S	PV-HKD 0202S	PV-HKD 02501S	PV-HKD 0301S	PV-HKD 0302S	PV-HKD 0303S	PV-HKD 00502S	PV-HKD 07503S
Standard reach	3/9 m												
Chain Dia(mm)	6.1	6.3	6.3	7.1	10	10	7.1	11.2	11.2	10	7.1	11.2	11.2
Lifting Speed(m/min)	7.1	6.8	3.4	6.6	8.8	6.6	3.3	5.4	5.4	4.4	2.2	2.7	1.8
Motor Power(kw)	0.75	0.75	0.75	1.5	3.0	3.0	1.5	3.0	3.0	3.0	1.5	3.0	3.0
Power Supply(V)	200-600												
ED Rating(%)	40	40	40	40	40	40	40	40	40	40	40	40	40
Chain falls	1	1	2	1	1	1	2	1	1	2	3	2	3



Type	Capacity (Ton)	Unit (mm)											
		H	A	B	D	E	I	J	K	L	M	N	Chain
PV-HKD00501S	0.5	540	455	240	285	165	φ 34	28	φ 34	25	19	19	φ 6.3
PV-HKD0102S	1	575	455	240	285	165	φ 34	28	φ 34	25	19	19	φ 6.3
PV-HKD0101S	1	650	520	260	300	176	φ 42	32	φ 42	32	24	24	φ 7.1
PV-HKD01501S	1.5	800	615	295	430	265	φ 49	40	φ 49	40	30	30	φ 10.0
PV-HKD0201S	2	800	615	295	430	265	φ 49	40	φ 49	40	30	30	φ 10.0
PV-HKD0202S	2	835	520	260	300	236	φ 49	40	φ 49	40	30	30	φ 7.1
PV-HKD02501S	2.5	845	615	295	430	265	φ 59	48	φ 59	48	35	35	φ 11.2
PV-HKD0301S	3	845	615	295	430	265	φ 59	48	φ 59	48	35	35	φ 11.2
PV-HKD0302S	3	950	615	295	430	320	φ 59	48	φ 59	48	35	35	φ 10.0
PV-HKD0303S	3	950	520	260	350	205	φ 59	48	φ 59	48	35	35	φ 7.1
PV-HKD0502S	5	1030	615	295	430	325	φ 60	48	φ 60	48	43	43	φ 11.2

Type	Capacity (Ton)	Unit (mm)											
		H	A	B	D	E	K	L	N	I	J	P	Chain
PV-HKDH00501S	0.5	555	455	240	285	165	φ 34	28	19	φ 26	115	221	φ 6.3
PV-HKDH0102S	1	630	455	240	285	165	φ 42	32	24	φ 26	115	221	φ 6.3
PV-HKDH0101S	1	635	520	260	300	176	φ 42	32	24	φ 26	115	221	φ 7.1
PV-HKDH01501S	1.5	764	615	295	430	265	φ 49	40	30	φ 31	138	265	φ 10.0
PV-HKDH0201S	2	764	615	295	430	265	φ 49	40	30	φ 31	138	265	φ 10.0
PV-HKDH0202S	2	800	520	260	300	236	φ 49	40	30	φ 31	138	265	φ 7.1
PV-HKDH02501S	2.5	854	615	295	430	265	φ 59	48	35	φ 36	157	277	φ 11.2
PV-HKDH0301S	3	854	615	295	430	265	φ 59	48	35	φ 36	157	277	φ 11.2
PV-HKDH0302S	3	960	615	295	430	320	φ 59	48	35	φ 36	157	277	φ 10.0
PV-HKDH0303S	3	968	520	260	350	205	φ 59	48	35	φ 36	157	277	φ 7.1
PV-HKDH0502S	5	1007	615	295	430	325	φ 60	48	43	φ 43	178	332	φ 11.2

Type	Capacity (Ton)	Unit (mm)													
		H	A	B	D	E	I	K	L	N	W	U	R	T	Chain
PV-HKDM00501S	0.5	610	455	240	285	165	φ 31	φ 34	25	19	206	111	142	231	φ 6.3
PV-HKDM0102S	1	665	455	240	285	165	φ 31	φ 42	32	24	206	111	142	231	φ 6.3
PV-HKDM0101S	1	650	520	260	300	176	φ 31	φ 42	32	24	206	111	142	231	φ 7.1
PV-HKDM01501S	1.5	770	615	295	430	265	φ 36	φ 49	40	30	237	127	142	231	φ 10.0
PV-HKDM0201S	2	770	615	295	430	265	φ 36	φ 49	40	30	237	127	142	231	φ 10.0
PV-HKDM0202S	2	815	520	260	300	236	φ 36	φ 49	40	30	237	127	142	231	φ 7.1
PV-HKDM02501S	2.5	830	615	295	430	265	φ 43	φ 59	48	35	265	140	142	231	φ 11.2
PV-HKDM0301S	3	830	615	295	430	265	φ 43	φ 59	48	35	265	140	142	231	φ 11.2
PV-HKDM0302S	3	930	615	295	430	320	φ 43	φ 59	48	35	265	140	142	231	φ 10.0
PV-HKDM0303S	3	940	520	260	350	205	φ 43	φ 59	48	35	265	140	142	231	φ 7.1
PV-HKDM0502S	5	1015	615	295	430	325	φ 54	φ 60	48	43	296	156	142	231	φ 11.2

2.5 Electric trolley

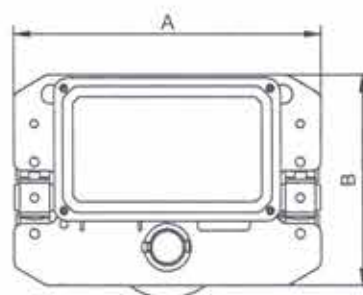
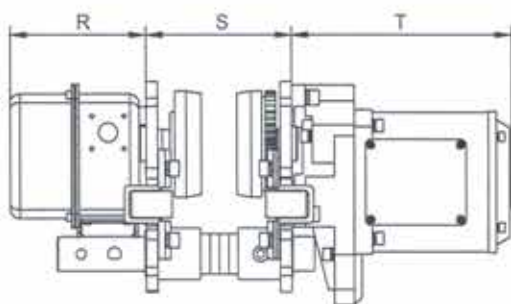


Table 2.5 Technical parameters-electric trolley

Type	Capacity (ton)	A	B	R	S I-beam (mm)	T	Speed (m/min) 50Hz	Motor (Kw)	Min. radius of turn	Net Weight (Kg)
PV-DDPC-0.5	0.5	315	212	142	58-153	231	11 or 21	0.4	0.8	45
PV-DDPC-01	1	315	212	142	58-153	231	11 or 21	0.4	0.8	45
PV-DDPC-02	2	325	220	142	82-178	231	11 or 21	0.4	0.8	53
PV-DDPC-03	3	340	250	142	100-178	231	11 or 21	0.75	1.0	65
PV-DDPC-05	5	400	291	142	100-178	231	11 or 21	0.75	1.8	88
PV-DDPC-7.5	7.5	400	291	142	100-178	230	11 or 21	0.75	1.8	98

PV-HKD	M	03	02	S	SD
1	2	3	4	5	6
1	Model				
2	Type	M=Electric Trolley			
		HC= Hand Chain Block			
		H= Manual Trolley			
		SL= Low Headroom			
3	Capacity				
4	No. Of Chain falls				
5	Lifting Speed	S= Single Speed			
		L= Single Speed of Low Lifting			
		SD= Dual Speed			
6	Operating Speed	H=High Speed			
		L=Low Speed			
		SD= Dual Speed			

3.Safety instructions



Danger

The hoist is not designed for lifting people. Any modification including increasing lifting speed or design changes must only be done by the original manufacturer or a qualified engineer.

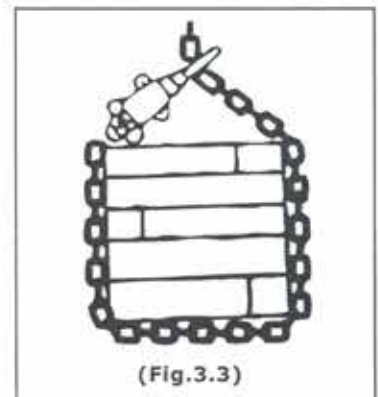
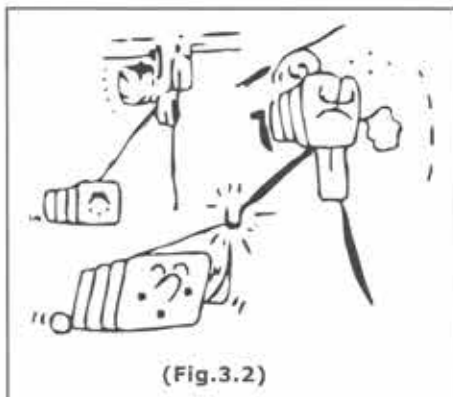
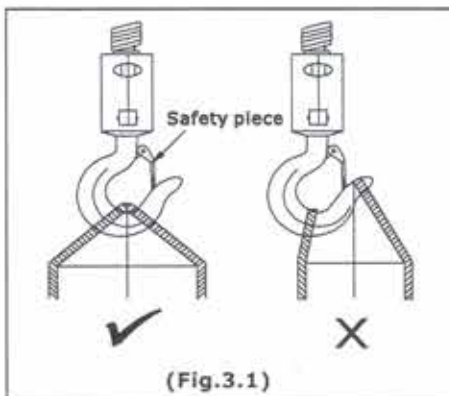


Danger

This hoist must never be operated in an explosive environment.

1. Only a competent person should operate this hoist.
2. Before operation confirm the following:
 - 2.1 The correct sling is used to attach the load to the lower hook.
 - 2.2 The sling must be attached to the lower hook as indicated in figure 3.1 and the safety latch must be securely in place.
 - 2.3 Make sure that the load to be lifted is safe and secure before beginning the lift.
3. The buttons on the control pendant must be pushed stably and firmly.
4. The operator must refrain from moving the load in small increments (inching).
5. The operator must ensure that the motor has stopped before operating in the reverse direction.
6. When lifting ensure that the load is evenly balanced. Never lift an unbalanced load.
7. Never drag a load along the ground as indicated in figure 3.2.
8. Never use the load chain as a sling as indicated in figure 3.3.
9. When in operation, the operator must always pay attention to the load and must never turn their back on the lifting operation.

10. Never lift loads that exceed the rated load capacity of the hoist.
 11. Never lift a load if the load chain is twisted or tangled.
 12. Regular inspection of the load chain is necessary to ensure the chain is in good condition.
- If the chain is damaged do not operate the hoist until this is replaced.



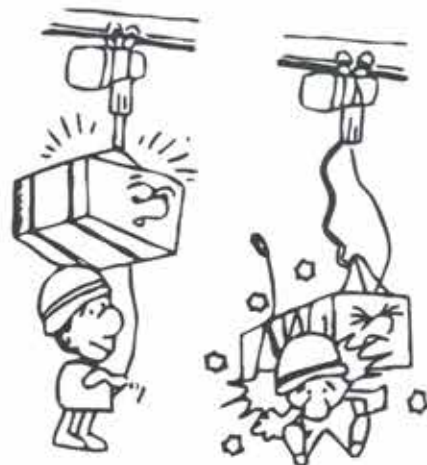
Warning

Never weld on or near the load chain



Warning

Never stand under the load when lifting



4. Installation procedures

4.1 Unpacking the electric chain hoist

After unpacking the hoist check to ensure that the control cable, gearbox and motor cover are not damaged. Every hoist should be equipped with the following accessories:

1.Chain bag	1
2.Control cable	3 meters (minimum)
3.Control pendent	1

4.2 Voltage



Note

If the power voltage is 10% above or under the standard voltage to operate the hoist, this will result in damage to the motor. Always confirm that the voltage supply is within the standard range prior to operation.

4.3 Installation of the electric chain hoist



Warning

Never connect the hoist's power supply prior to the completion of the installation.

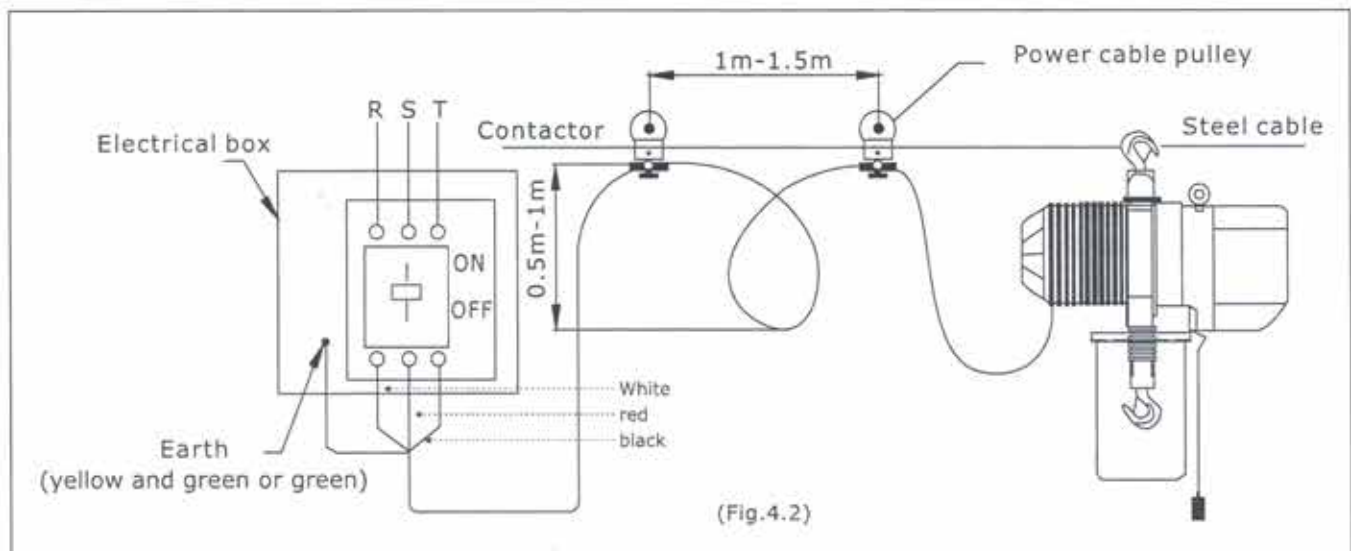
4.3.1 Before installing the hoist ensure that the upper hook assembly has been firmly attached to the hoist body and that the load chain pin is properly fitted.

4.3.2 Attach the chain bag as indicated in figure 4.1

4.3.3 Connect the power supply and control pendant as indicated in figure 4.2.
This should only be completed by a competent person.



(Fig.4.1)



(Fig.4.2)

4.3.4 Operation test

- press the Ⓚ button to lower the hoist hook until the limit spring touches the limit switch and the motor stops automatically.
- press the Ⓚ button until the load chain is fully in the chain bag and the motor stops automatically.
- Test the function of the emergency stop switch (if supplied with the hoist) by pressing either the Ⓚ or the Ⓚ buttons and the emergency button at the same time. Once pressed the hoist must stop and the buttons on the control pendant should no longer be functional. Turn the emergency switch clockwise to release and the hoist should be operational again.
- Check to see that the welds on the load chain all face the same direction as indicated in figure 4.3. The hoist will not operate properly if this is not correct.

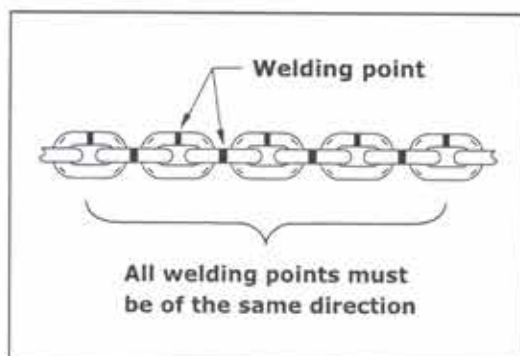


Fig.4.3

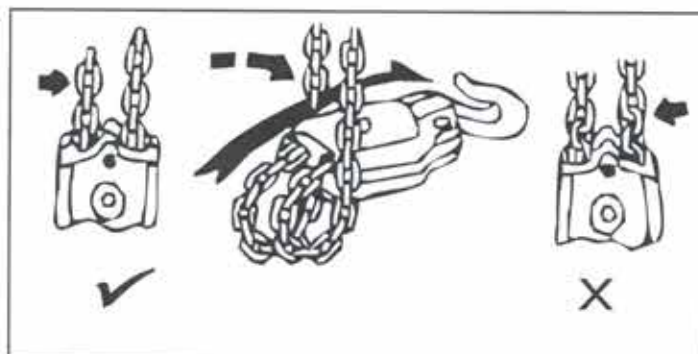


Fig.4.4



Note

If the hoist is a double fall unit make sure that the load chain is not twisted and that the lower hook has not been reversed as indicated in figure 4.4.

Fig.4.4

e) Chain lubrication

The degree of lubrication applied to the load chain plays an important role in prolonging the service life of the chain. The operator must regularly apply oil to the load chain.

1. Keep the load chain vertical without a load
2. Clean the load chain of any dust or rust.
3. Lubricate each link and run the load chain up and down without a load to evenly distribute the lubrication.

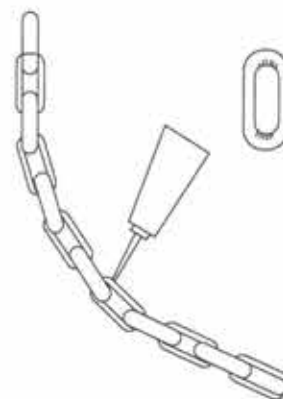
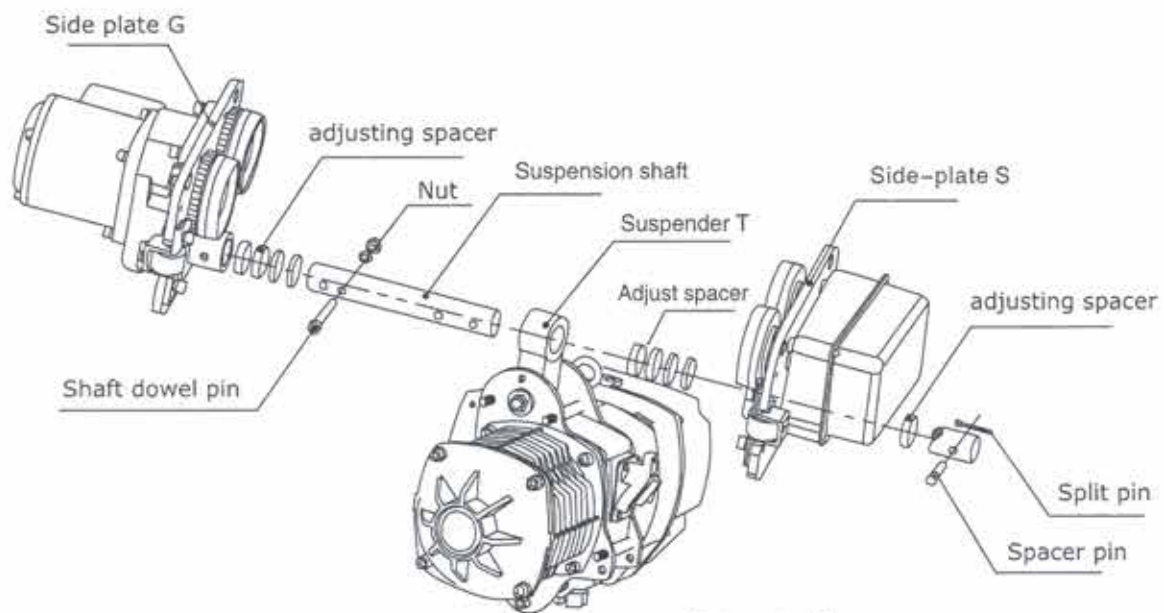


Fig.4.5

4.4 INSTALLATION OF THE ELECTRIC CHAIN HOIST AND ELECTRIC TROLLEY

a) Installation of trolley

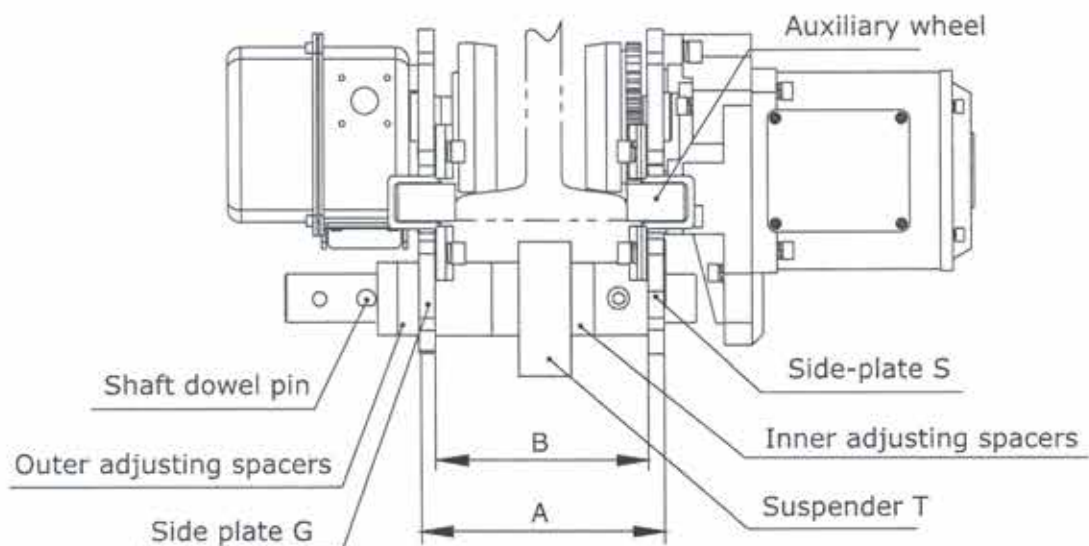
- 1) Insert the suspension shaft into the side plate G and fix them together with a bolt and nut.
- 2) Insert the suspension shaft and adjusting spacer through the suspender T. The total width should be about 1/2 the width of the traversing beam.
- 3) Fit the side plate S between the trolley wheel and the electric case.
- 4) Used the shaft dowel pin and split pin to secure the hoist.



(Fig.4.6)

b) Adjusting the trolley width

- 1) Use the spacers to adjust the B dimension and ensure that the wheels on the side plates run snugly on the beam.
- 2) To disassemble the trolley first remove the shaft dowel pin and follow the installation instructions in reverse.



(Fig.4.7)



Warning

Never connect the power supply before the installation is completed.

c) Installing the trolley on the traversing beam

- 1) For greater convenience always install the trolley from the end of the beam.
- 2) If this is not possible follow these steps:
Remove the bolt from the suspension shaft hole A. Then insert it into the hole B and fix with the split pin. This ensures the trolley won't fall apart.
Open side plate S away from side plate G so that the trolley can fit over the traversing beam.
Once fitted to the beam push both side plates together so that the wheels of the trolley fit snugly onto the traversing beam.
Remove the bolt from the suspension shaft hole B. Then insert it into the hole A and fix with the split pin.

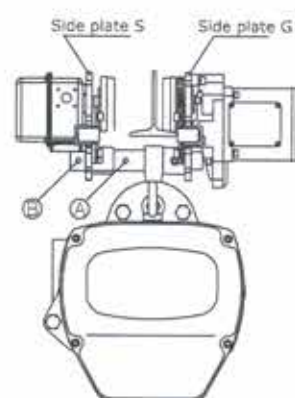


Fig 4.8
Trolley with the hoist
mounting to traversing rail

5. Operating procedures

Once the hoist has been correctly assembled and tested by a competent person it may be used. The operator must have a clear and unobstructed view of the whole working area and must check to ensure this is safe. When the trolley is operated, the operator must be careful not to cause the load to swing in the opposite direction to the travel of the trolley when changing direction.



Warning

Lifting a heavy load is a dangerous operation with unexpected risks. Follow all the safety instructions in chapter 3 and pay attention to the following.

- (1) Before operation, the operator must have a clear and unobstructed view of the whole working area.
- (2) Before operation, the operator must check whether the whole working area is safe and free of risks.
- (3) When operating the trolley, be careful to prevent the load swinging when changing the direction of movement.

6. Maintenance and inspection



Danger

Never perform maintenance or inspections on the hoist while under load except when performing an inspection on the brake or limit switch.



Danger

Before maintenance on the hoist ensure that the power supply and control pendant are labelled "danger" and "inspection".

6.1 Maintenance

1. When the hoist operating time exceeds 500 hours check the lubrication of the unit and then regularly every 3 months thereafter using engine oil of the same grade as ISOVG46.
2. Ensure the hoist is clean and dry. Abuse of the hoist will reduce the durability of the unit.
3. If the hoist is operated outdoors, make use of a protection cover.

6.2 Inspection

1. Daily inspection before operation, check the following items:
 - a) the power supply
 - b) test the up, down and emergency stop controls without a load on the hoist
 - c) the motor is working normally
 - d) there is no excessive or unusually high noise
 - e) the safety latches fitted to the hooks are fitted properly and work normally
 - f) the movable parts of the limit switch and brake function normally
 - g) the chain is well lubricated

(2) Monthly inspection



Warning

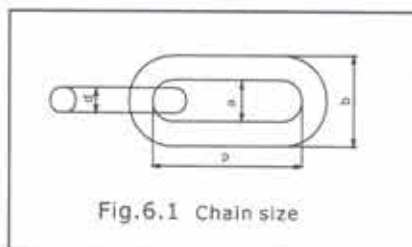
Only use original parts recommend by the manufacturer

a) Chain

Any distortion, elongation or wear of the chain may result in the chain not fitting the chain wheel. To ensure safe operation, the internal length and width and external width of the chain must be checked monthly.

- 1) The internal length (p) of the chain is greater than 5% above the figure in Table 6.1
- 2) The diameter (d) is less than 10% of the figure listed in Table 6.1

If the wear on the chain is high, check the chain wheel and guide for excessive wear as well.

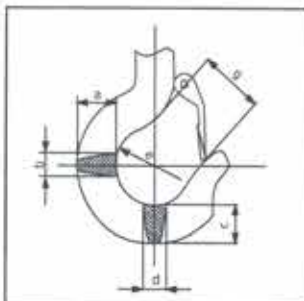


Diameter (mm) (d)	Capacity (ton)	Internal length (mm) (p)	Internal width (mm) (a)	External width (mm) (b)
6.3	0.3, 0.5	19	7.5	20.5
7.1	1, 2, 3	21	8.9	23
10.0	1.5, 2, 3	30	12.5	33
11.2	2.5-7.5	34	14	37.2

Table 6.1

b) Hoist hook

Check the hook for cracks or distortion. If the dimensions exceed 5% of the figures in Table 6.2 it should be replaced.



Capacity (ton)	a	b	c	d	e	g
0.5	27	18	25	17	35	28
1	34	24	30	24	42	32
2	46	29	39	30	49	40
3	56	35	49	34	59	48
5	67	43	57	44	60	48
7.5	82	55	80	48	85	80

Table 6.2

(C) Limit switch

Check the limit switch to ensure it operates normally and after cleaning cover with a thin layer of lubrication. The limit switch should only be inspected by a qualified electrician.

(3) Annual inspection

- Check whether there is excessive wear or damage to the gears
- Replace the oil in the gearbox
- Once checked the machine must be reassembled and tested prior to restarting operations.
Testing of the hoist should only be conducted by a competent person.

Specification of chain bag

Serial number of chain bag	Chain specifications	Chain length(m)	Size of chain bag(mm)	Material
8#	∅ 6.3	<6	150 x 150 x 350	canvas
8#	∅ 7.1	<8.8	150 x 150 x 350	canvas
12#	∅ 10.0, ∅ 11.2	<10	215 x 215 x 410	canvas
14#	∅ 10.0, ∅ 11.2	10-20	215 x 215 x 500	canvas
Other	∅ 6.3, ∅ 7.1 ∅ 10.0, ∅ 11.2	>20	/	Steel

Measuring the chain link wear using the gauge

- Measure the interlink distance as shown in Fig 6.3
- Measure the diameter of each link as shown in Fig 6.4 and replace the entire length of chain if the diameter is reduced by more than 10%
- Worn, distorted or elongated chain may result in the failure of a link
- Only a competent person should replace the chain
- If you are replacing the chain for a second time then you should also replace the chain transmission core shaft, adjuster and guide wheel
- The chain must be free of nicks, gouges, pitting or corrosion
- Check that there are no foreign objects between the chain links

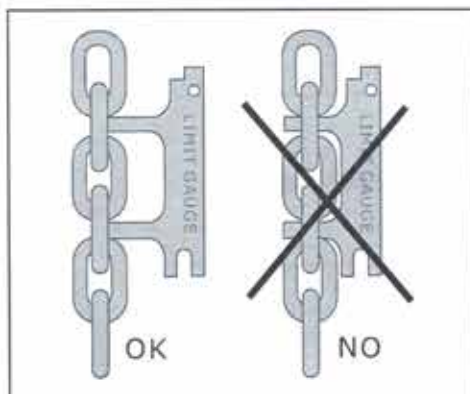


Fig.6.3 Measure of internodal distance

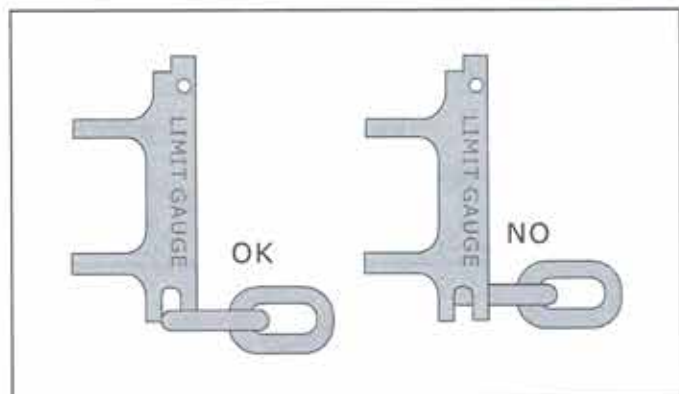


Fig.6.4 Measure of diameter

7. Wiring procedures

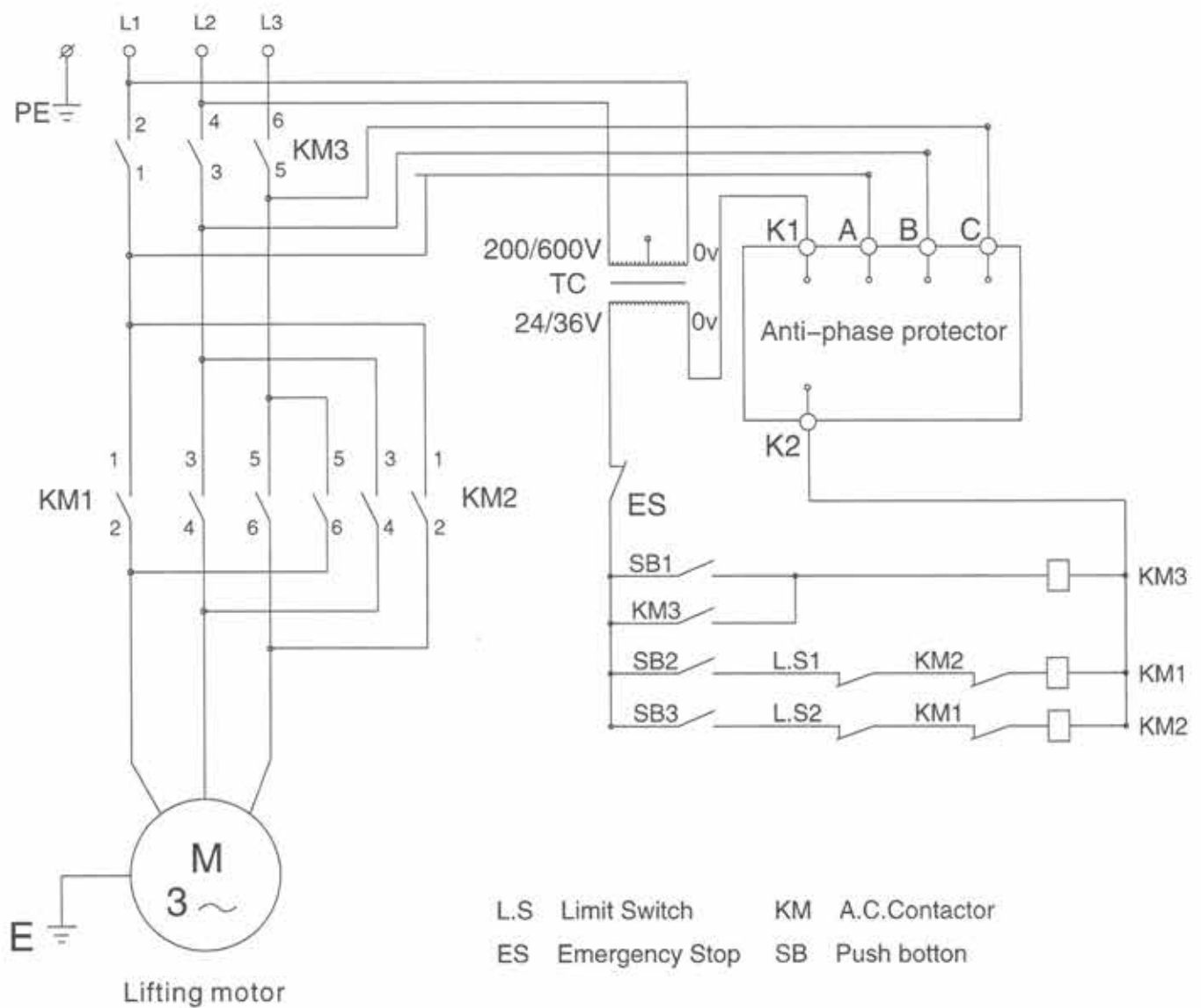
(1) 2 Points button switch single speed wiring diagram 200–600V(24/36/48V)(with emergency stop)	19
(2) 4 Points button switch single speed wiring diagram 200–600V(24/36/48V)(with emergency stop)	20
(3) 6 Points button switch single speed wiring diagram 200–600V(24/36/48V)(with emergency stop)	21
(4) 2 Points button switch single speed wiring diagram 200–600V(24/36/48V)(with emergency stop and overload protector)	22
(5) 4 Points button switch single speed wiring diagram 200–600V(24/36/48V)(with emergency stop and overload protector)	23
(6) 6 Points button switch single speed wiring diagram 200–600V(24/36/48V)(with emergency stop and overload protector)	24
(7) 2 Points button switch dual speed wiring diagram 200–600V(24/36/48V)(with emergency stop)	25
(8) 4 Points button switch dual speed wiring diagram 200–600V(24/36/48V)(with emergency stop)	26
(9) 4 Points button switch dual speed wiring diagram 200–600V(24/36/48V)(with emergency stop and overload protector)	27

Electric specifications are as follow:

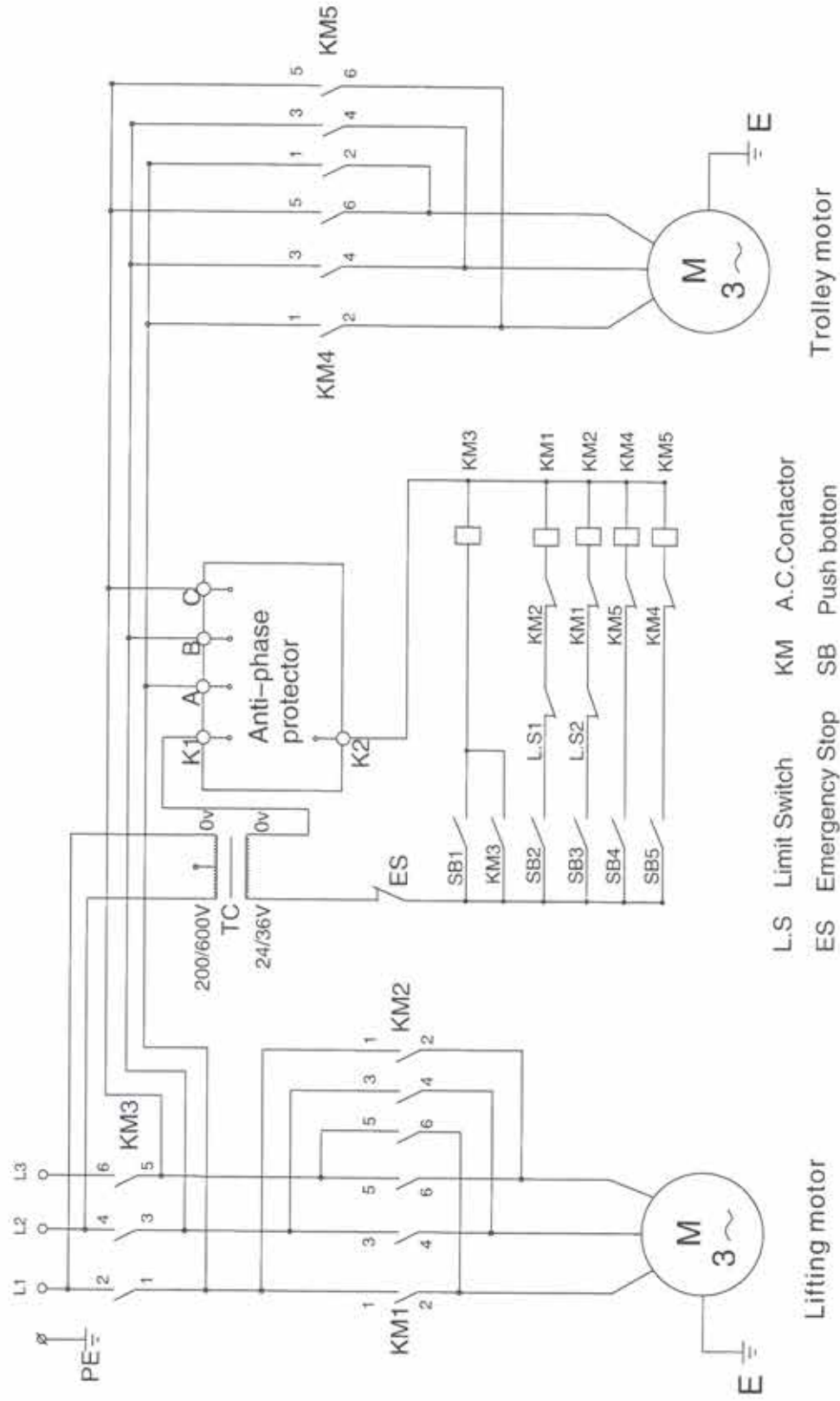
- (a) 3 phases
- (b) Frequency:50Hz/60Hz
- (c) Single voltage or dual voltage

Voltage Frequency	Dual voltage	Single voltage
	220V/380V 220V/440V 230V/460V	220V~600V
50Hz		
60Hz		

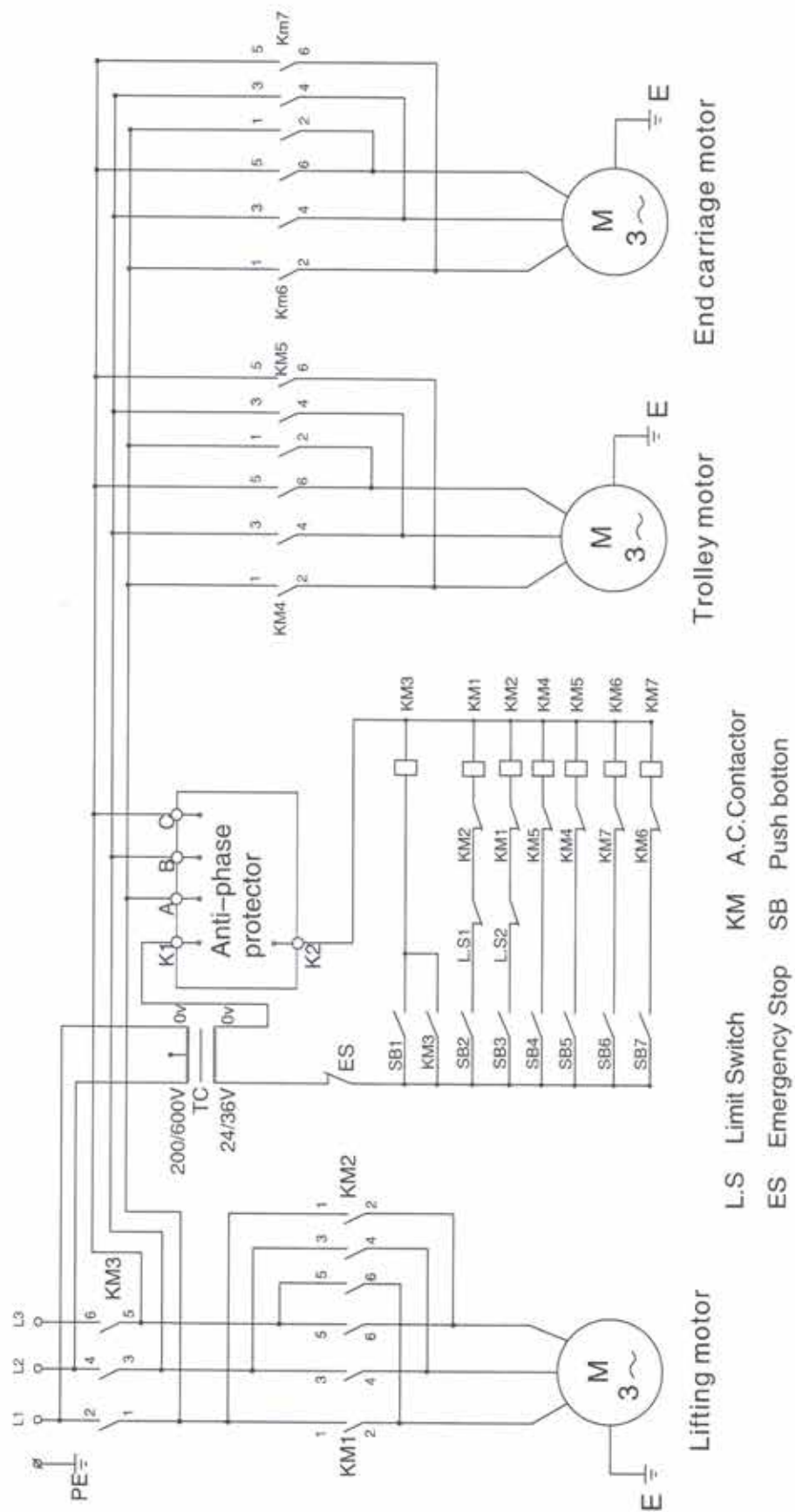
2 Points single speed (With emergency stop)



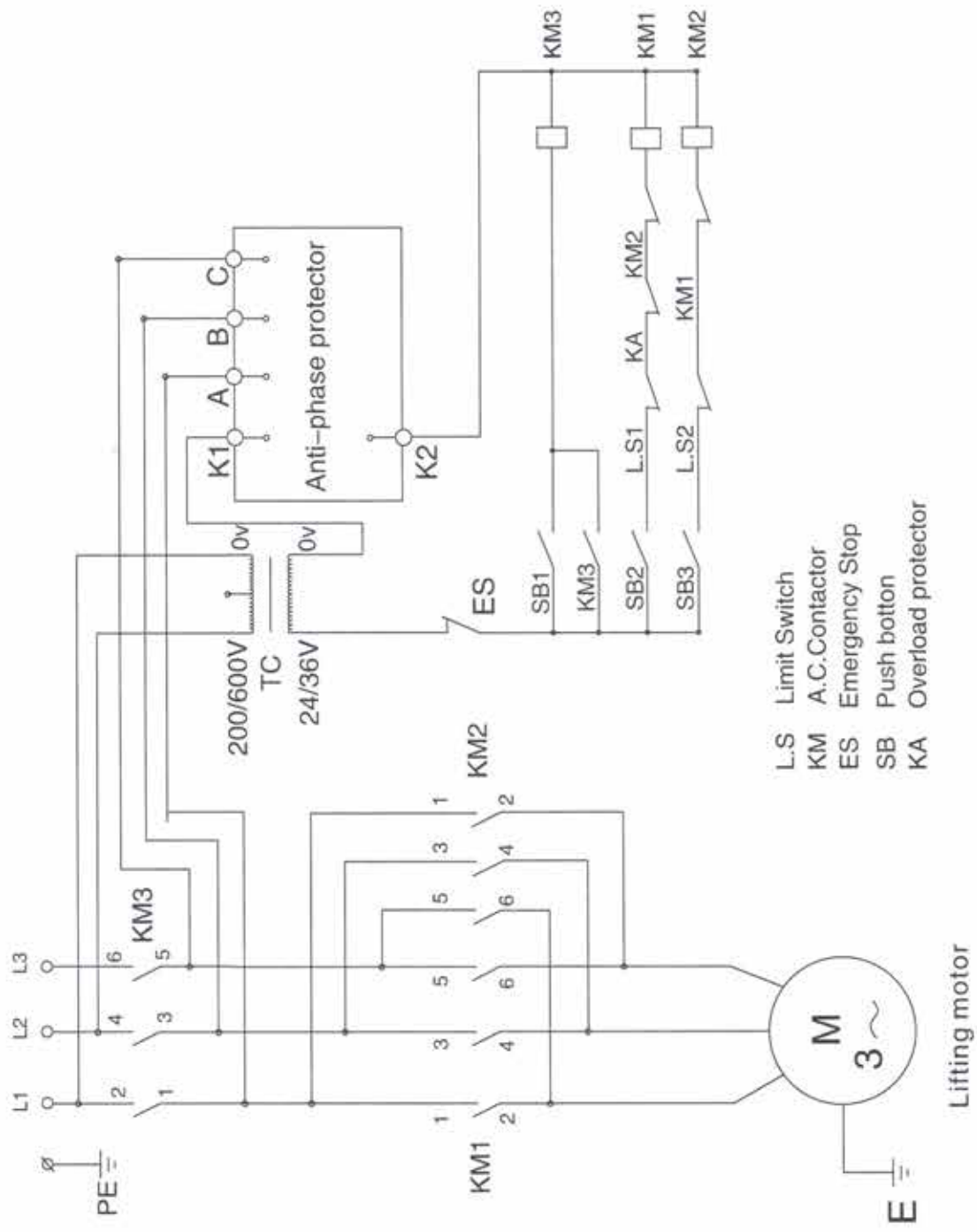
4 Points single speed (With emergency stop)



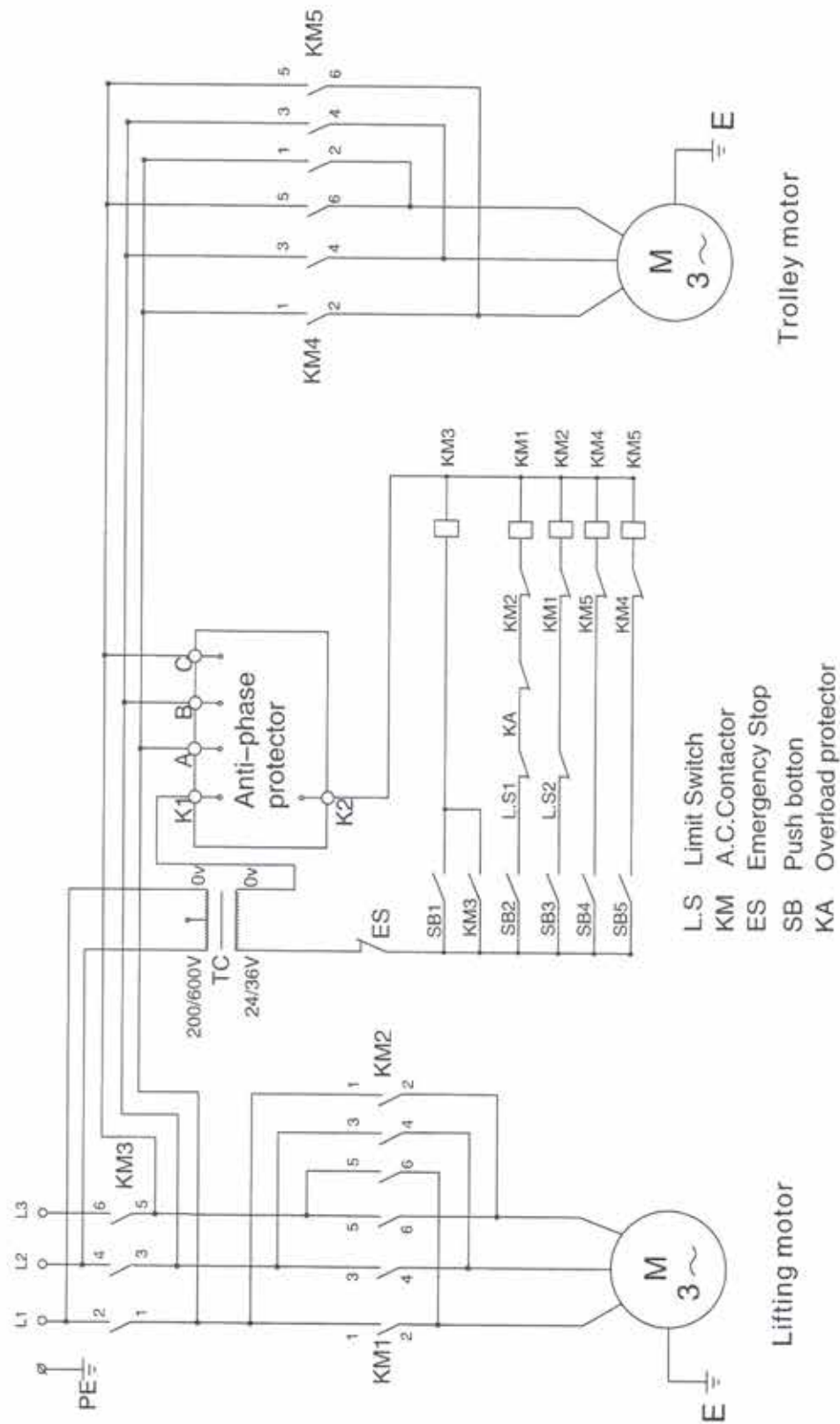
6 Points single speed (With emergency stop)



2 Points single speed (With emergency stop and overload protector)

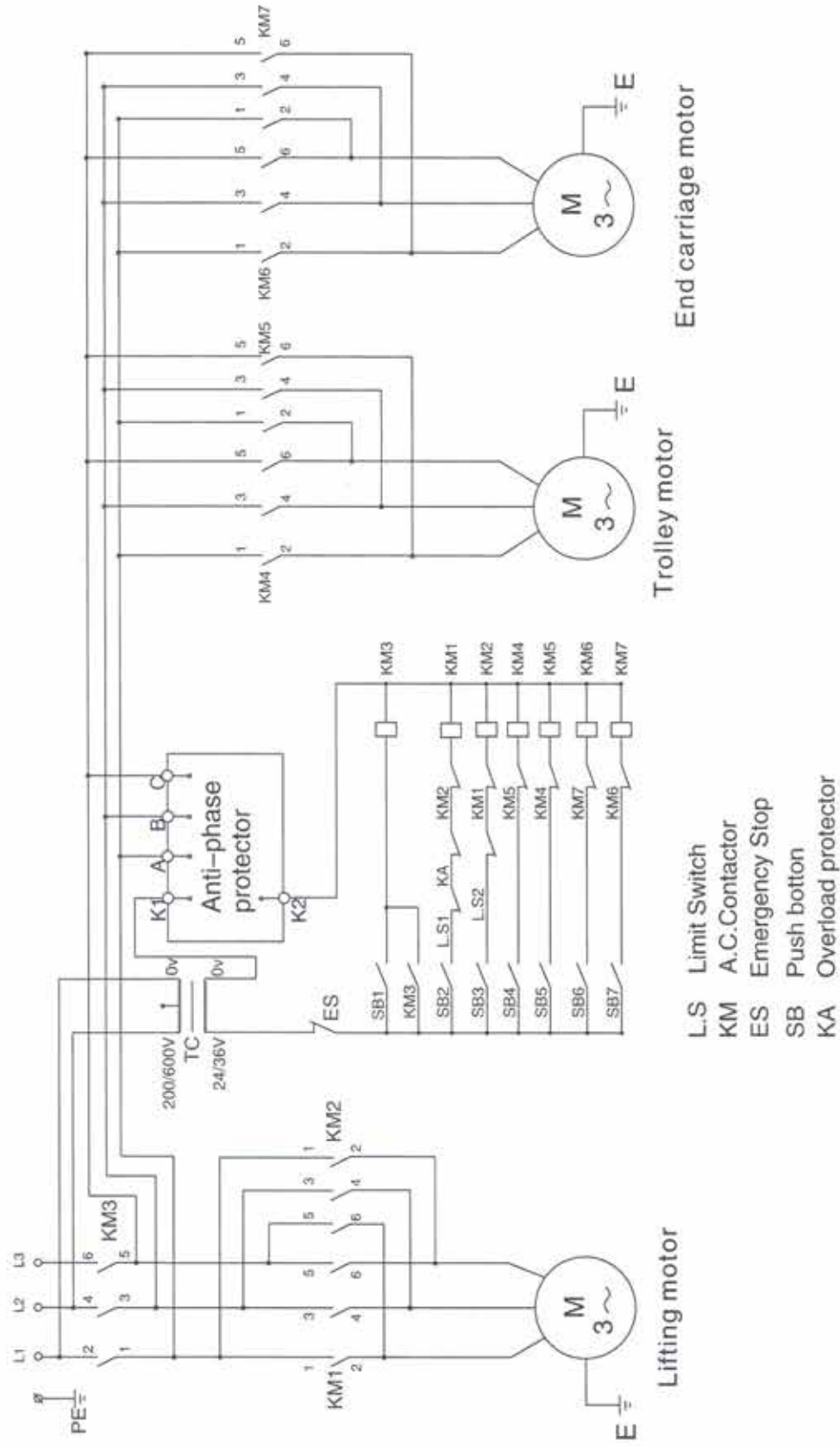


4 Points single speed (With emergency stop and overload protector)

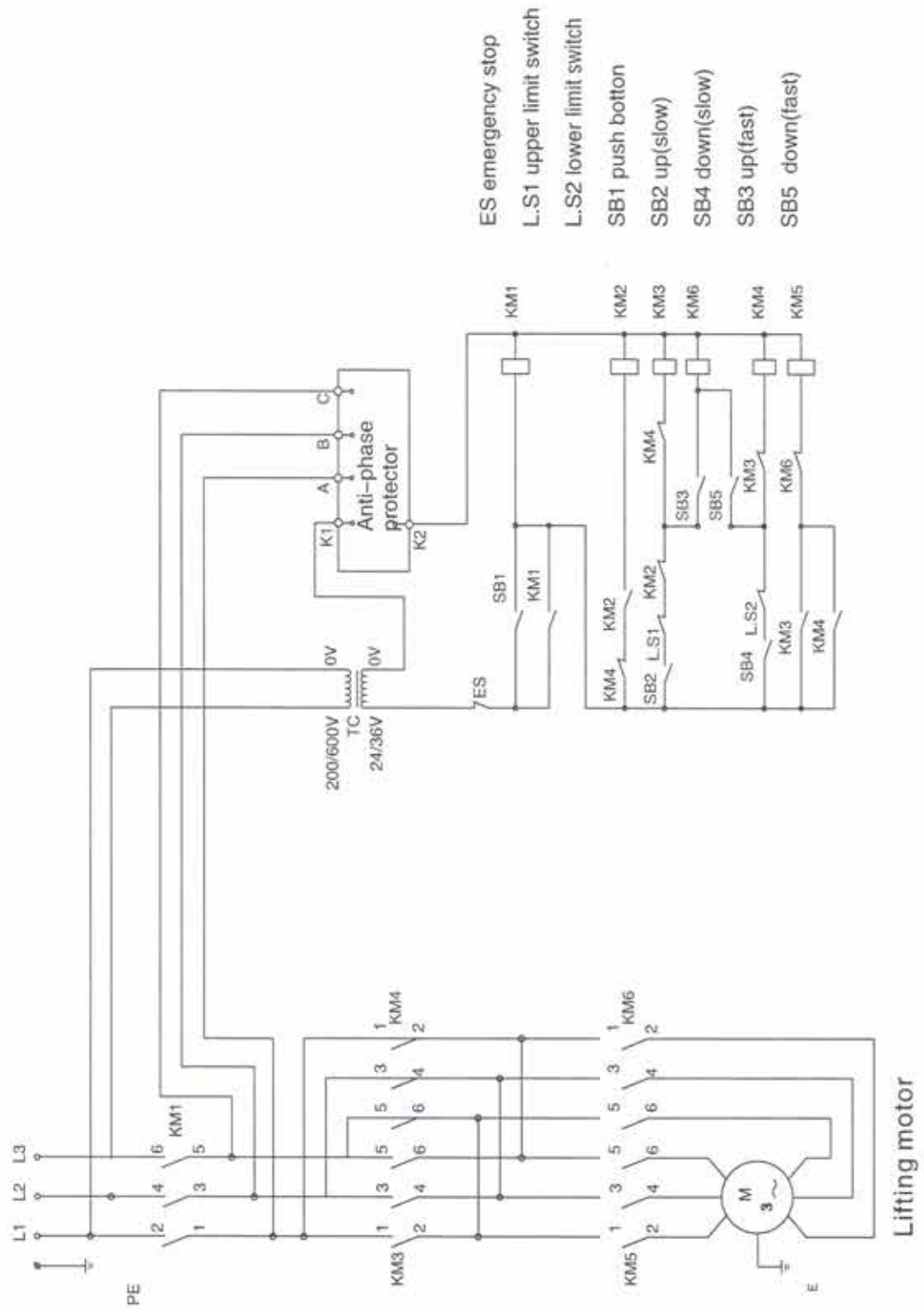


6 Points single speed

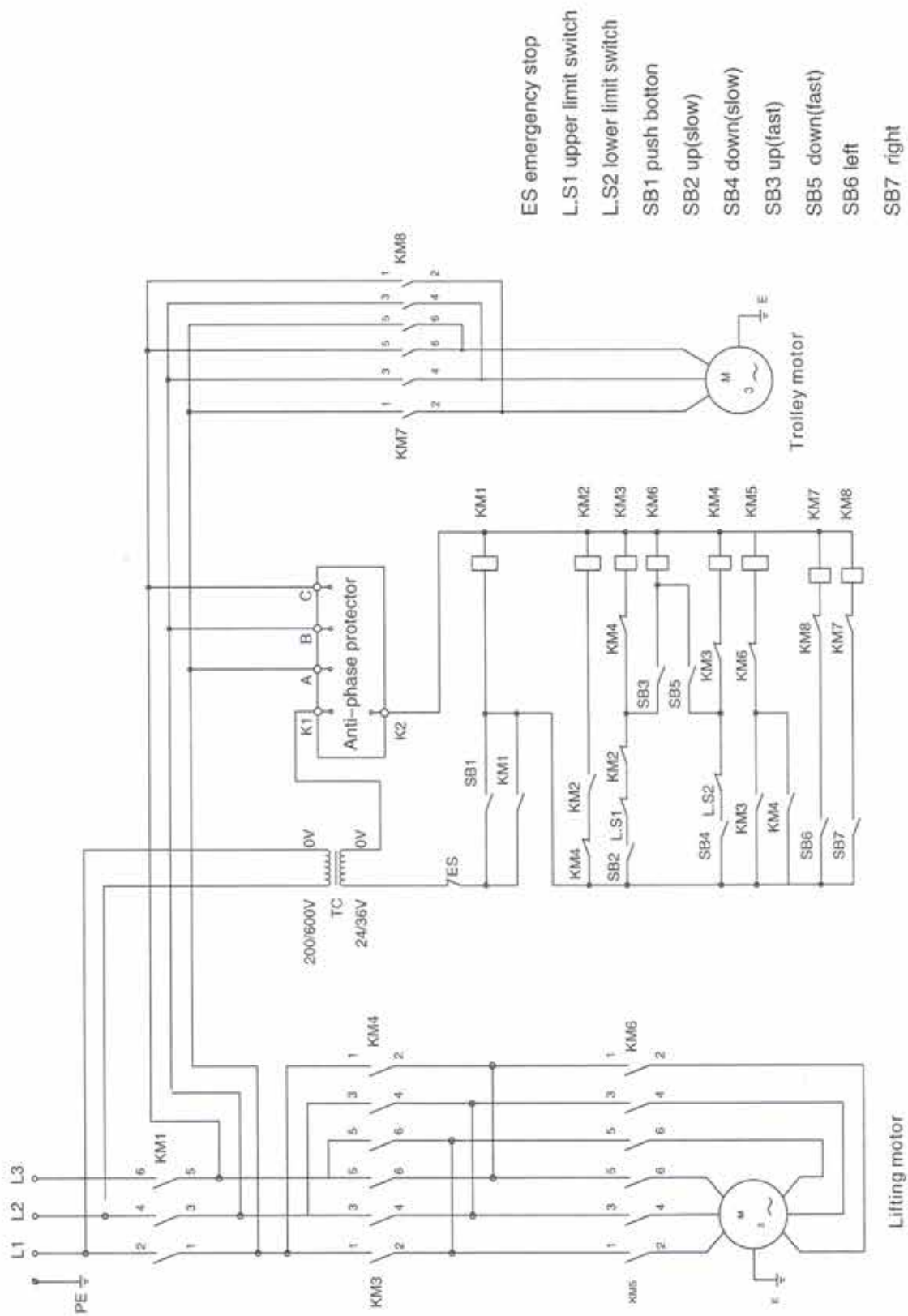
(With emergency stop and overload protector)



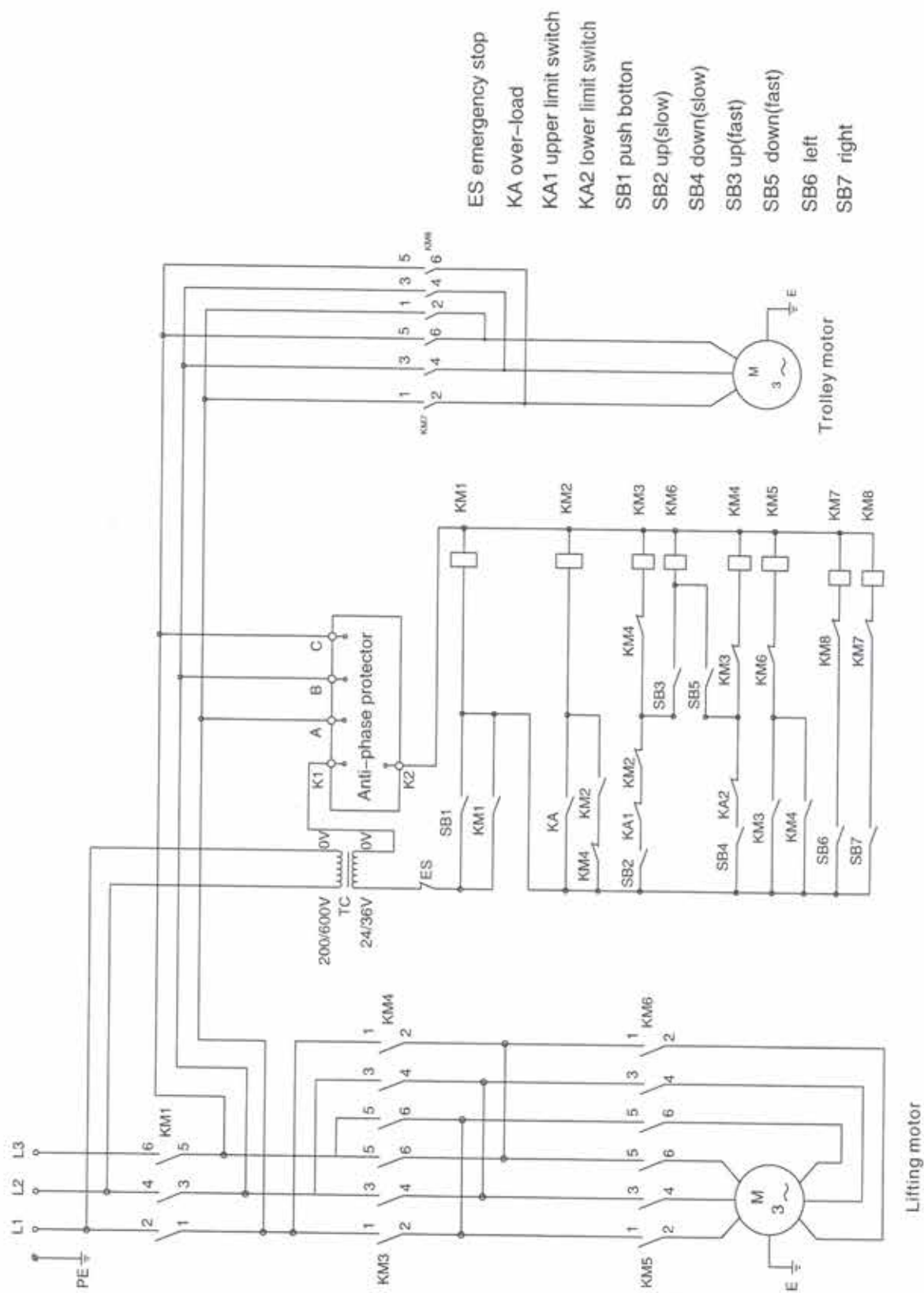
2 Points dual speed (With emergency stop)



4 Points dual speed (With emergency stop)



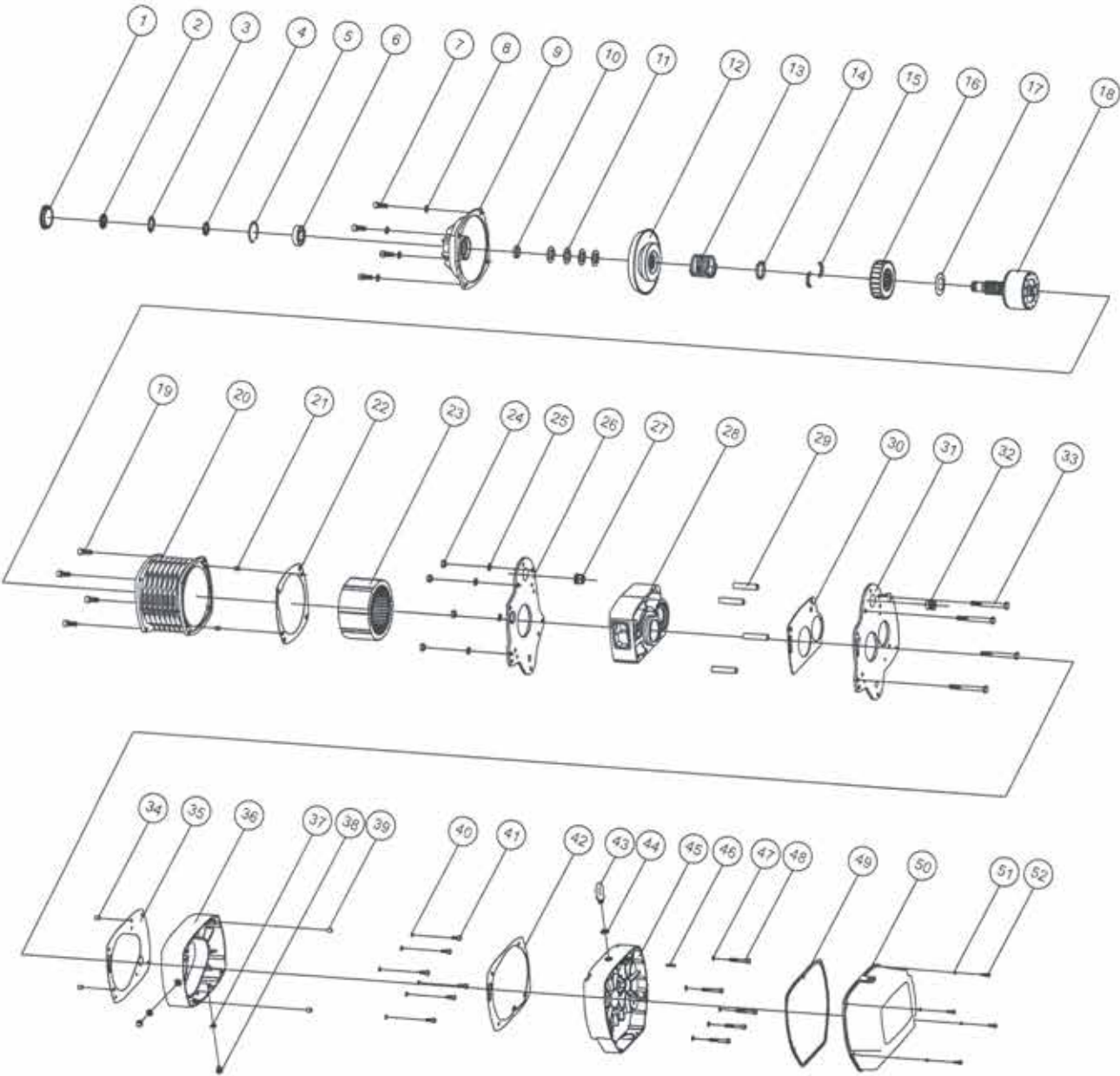
4 Points dual speed (With emergency stop and overload protector)



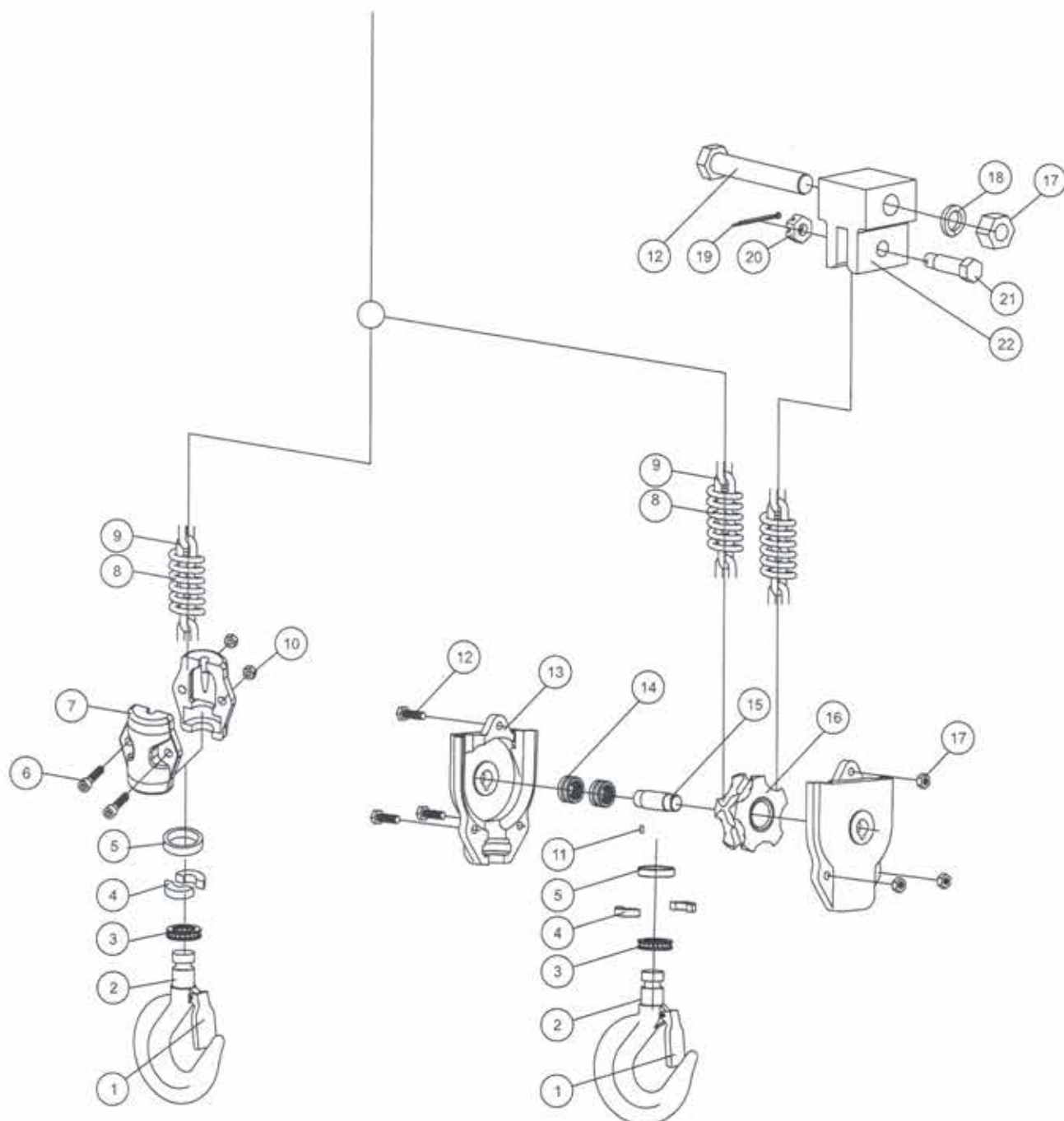
8. Diagrams and parts lists

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VIII-1 Motor and body assembly drawing



VIII-2 A HOOK ASSEMBLY DRAWING (0.5,1,1.5,2,2.5,3,5TON)

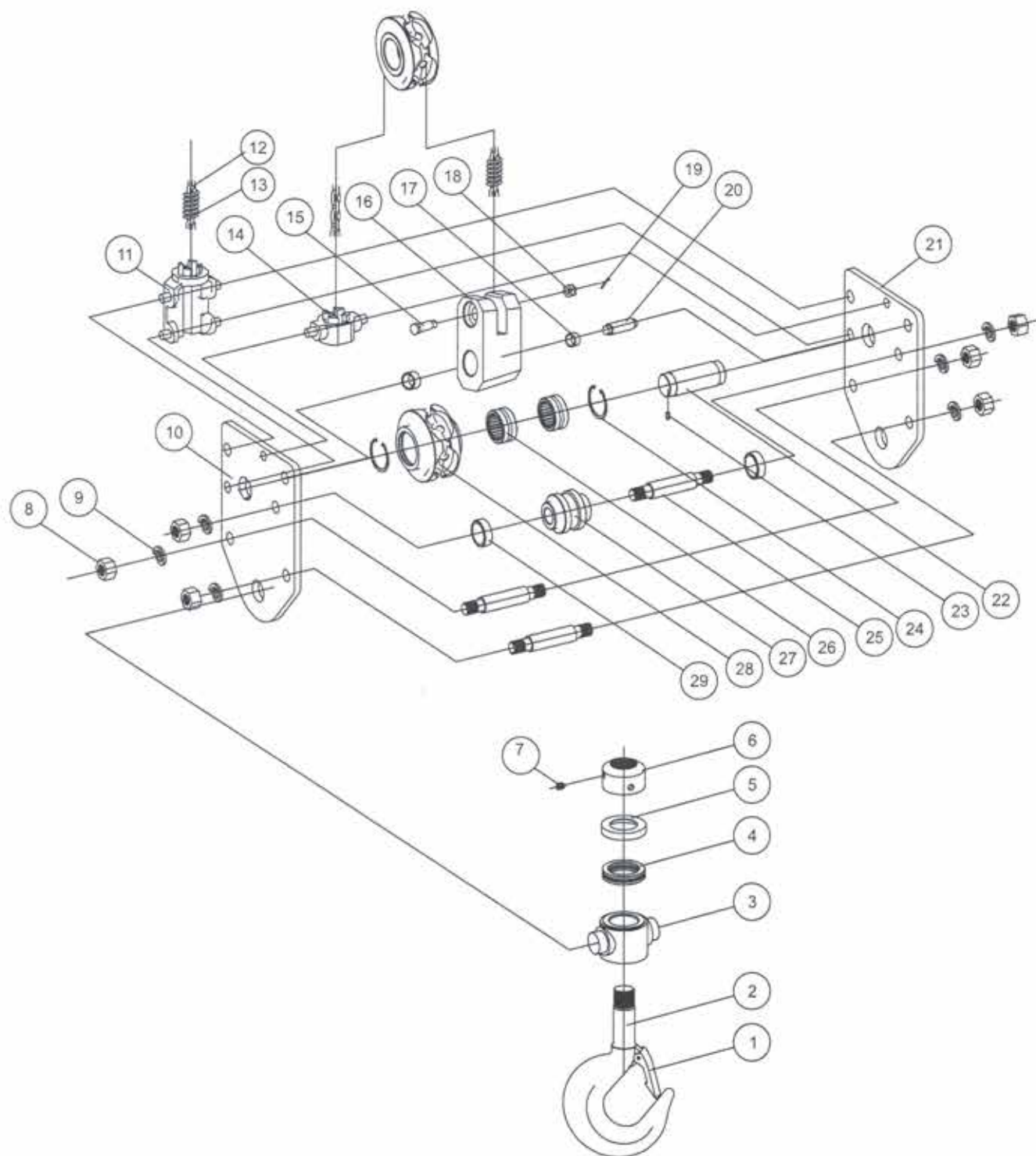


VIII-2A HOOK PARTS LIST(0.5,1,1.5,2,2.5,3,5TON)

S = SINGLE SPEED ; D = DUAL SPEEDS

Item	Spare Parts Name	Every Type Hoist Use Quantity																	
		00301,00501		0101		01501		0201		0202		02501		0301		0302		0502	
		S	D	S	D	S	D	S	D	S	D	S	D	S	D	S	D	S	D
1	Hook safety latch	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	Lower hook	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	Thrust ball bearing	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	Lower hook split ring	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
5	Lower hook locking sleeve	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	Socket head cap screw	2	2	2	2	2	2	2	2	/	/	2	2	2	2	/	/	/	/
7	Lower hook swivel shell	2	2	2	2	2	2	2	2	/	/	2	2	2	2	/	/	/	/
8	Chain limit spring	1	1	1	1	1	1	1	1	2	2	1	1	1	1	2	2	2	2
9	Chain	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m
10	Self-locking nut	2	2	2	2	2	2	2	2	/	/	2	2	2	2	/	/	/	/
11	Slotted spring pins	/	/	/	/	/	/	/	/	1	1	/	/	/	/	1	1	1	1
12	Outer hexagonal shell	/	/	/	/	/	/	/	/	4	4	/	/	/	/	4	4	4	4
13	Lower load block shell	/	/	/	/	/	/	/	/	2	2	/	/	/	/	2	2	2	2
14	Needle bearing	/	/	/	/	/	/	/	/	2	2	/	/	/	/	2	2	2	2
15	Lower load block chain wheel shaft	/	/	/	/	/	/	/	/	1	1	/	/	/	/	1	1	1	1
16	Lower load chain wheel	/	/	/	/	/	/	/	/	1	1	/	/	/	/	1	1	1	1
17	Hexnut	/	/	/	/	/	/	/	/	4	4	/	/	/	/	4	4	4	4
18	Spring washer	/	/	/	/	/	/	/	/	1	1	/	/	/	/	1	1	1	1
19	Split pin	/	/	/	/	/	/	/	/	1	1	/	/	/	/	1	1	1	1
20	Hex slotted nut	/	/	/	/	/	/	/	/	1	1	/	/	/	/	1	1	1	1
21	Chain bolt	/	/	/	/	/	/	/	/	1	1	/	/	/	/	1	1	1	1
22	Chain locking block	/	/	/	/	/	/	/	/	1	1	/	/	/	/	1	1	1	1

VII-2B HOOK ASSEMBLY DRAWING(7.5TON)

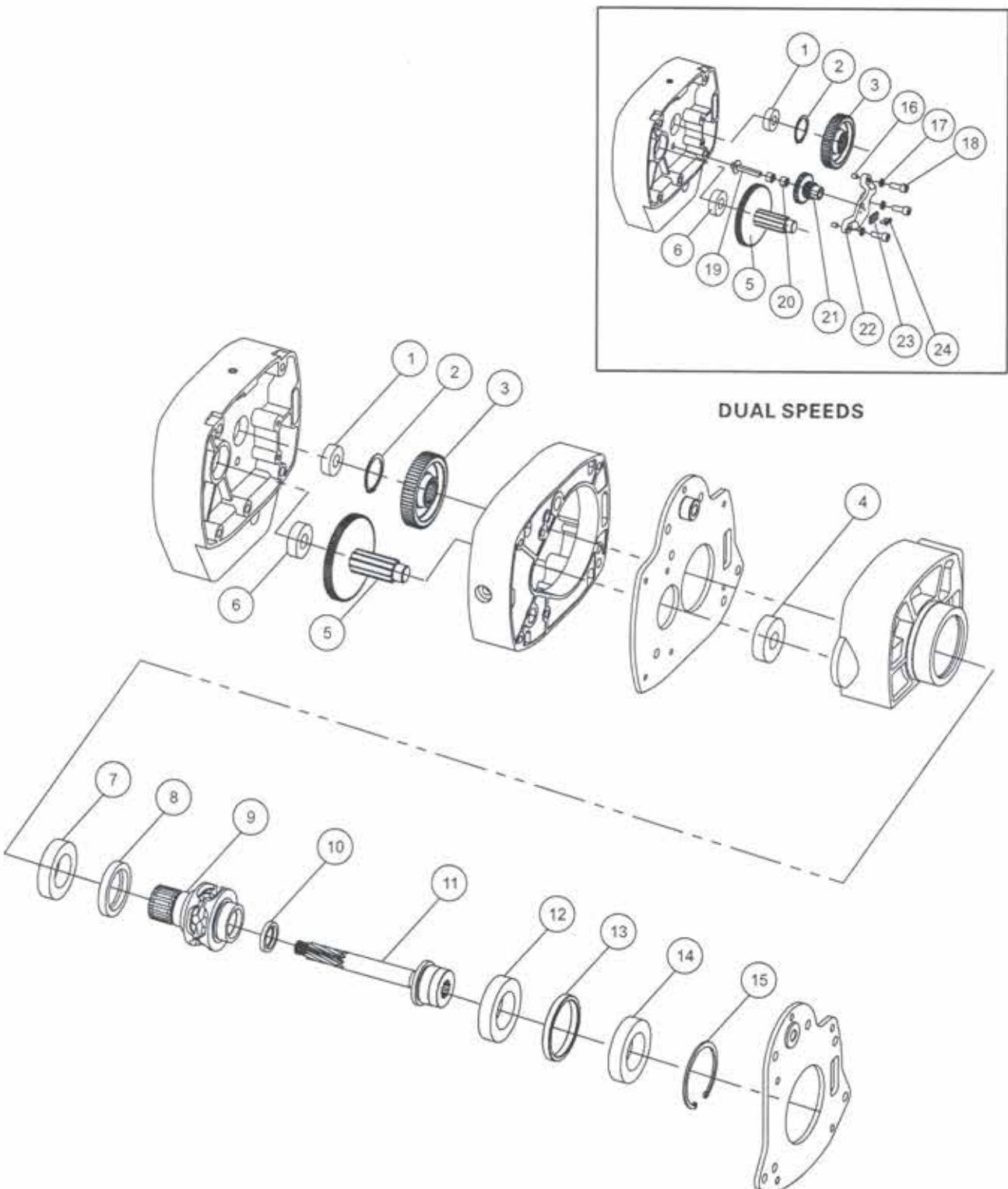


VII-2B HOOK ASSEMBLY PARTS LIST(7.5TON)

S = SINGLE SPEED ; D = DUAL SPEEDS

Item	Spare Parts Name	Every Type Hoist Use Quantity	
		0303,07503	
		S	D
1	Hook safety latch	1	1
2	lower hook	1	1
3	Hook to undertake block	1	1
4	thrust ball bearing	1	1
5	Bearing sheath	1	1
6	Hook nut	1	1
7	Hex--end tight set screws	1	1
8	Hexnut	6	6
9	Spring washer	6	6
10	Stent motherboard under	1	1
11	Chain guiding for inlet and outlet	1	1
12	Chain	9m	9m
13	Limit spring	1	1
14	Guide chain body	1	1
15	Chain bolt	1	1
16	Chain locking block	1	1
17	Locking block of gasket	2	2
18	Hex slotted nut	1	1
19	Split pin	1	1
20	Locking block of shaft	1	1
21	Stent vice board under	1	1
22	lower hook chain wheel shaft	1	1
23	Spring straight pins	1	1
24	Circlip for hole	2	2
25	Stents double nut	3	3
26	Needle bearing	2	2
27	Guiding pulley	1	1
28	Chain wheel	1	1
29	Circlip	2	2

VII-3 GEAR BOX ASSEMBLY DRAWING

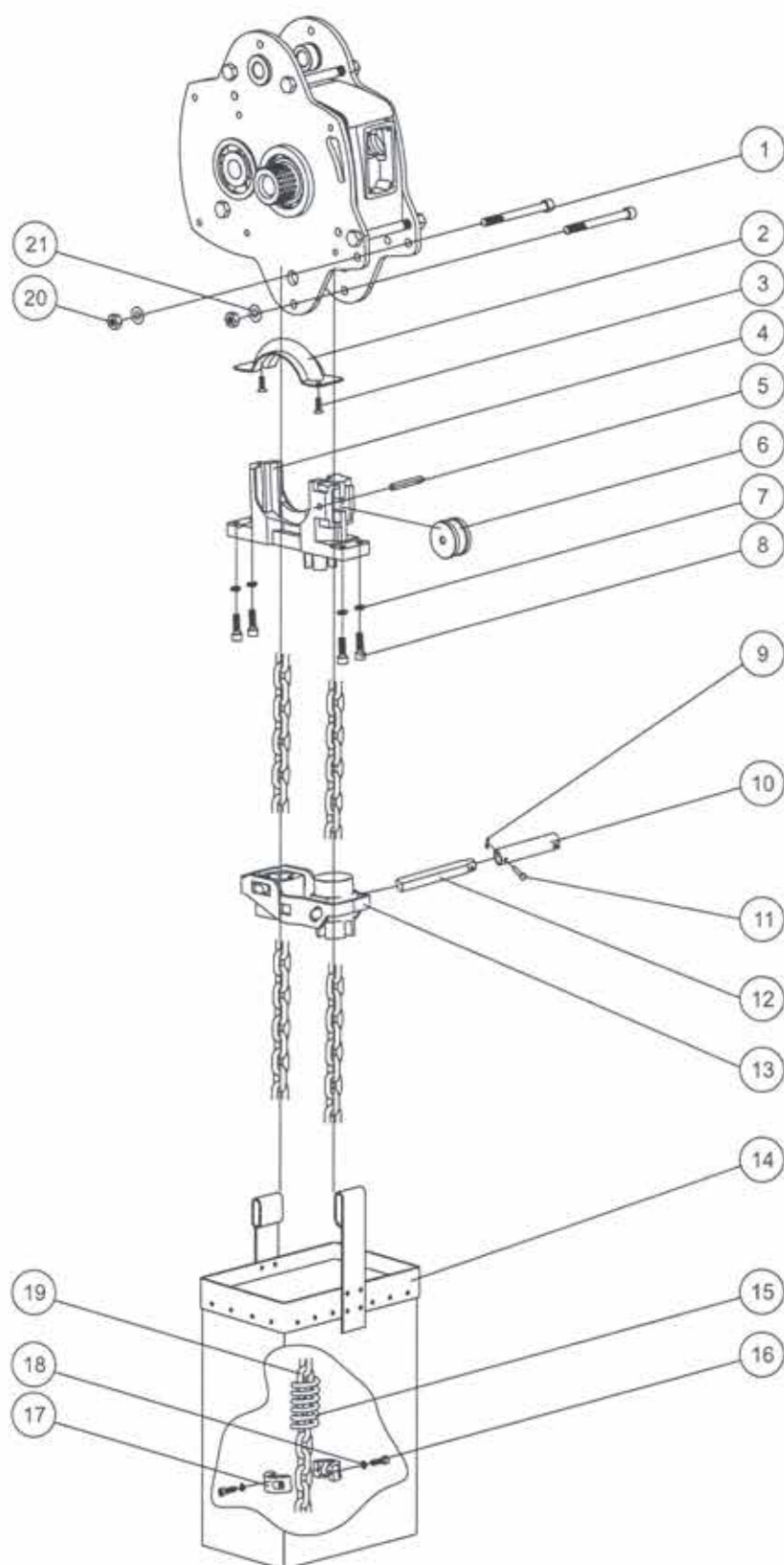


VII-3 GEAR BOX ASSEMBLY PARTS LIST

S = SINGLE SPEED ; D = DUAL SPEEDS

Item	Spare Parts Name	Every Type Hoist Use Quantity															
		00301,00501		0101		01501		0201		02501		0301		0502		07503	
		S	D	S	D	S	D	S	D	S	D	S	D	S	D	S	D
1	Deep groove ball bearing	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	circlip for shaft	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	Output gear	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	Deep groove ball bearing	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	Gear-gear shaft	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	Deep groove ball bearing	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	Deep groove ball bearing	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
8	Oil seal	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
9	Chain wheel	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
10	Oil seal	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
11	Output shaft assembly	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
12	Deep groove ball bearing	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
13	Bearings fixed ring	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
14	Deep groove ball bearing	1	1	1	1	/	/	/	/	/	/	/	/	/	/	/	/
15	Circlip for hole	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
16	Spring straight pins	/	2	/	2	/	2	/	2	/	2	/	2	/	2	/	2
17	Spring washer	/	3	/	3	/	3	/	3	/	3	/	3	/	3	/	3
18	socket head cap screws	/	3	/	3	/	3	/	3	/	3	/	3	/	3	/	3
19	Reduction gear fixed axis	/	1	/	1	/	1	/	1	/	1	/	1	/	1	/	1
20	Bearing	/	2	/	2	/	2	/	2	/	2	/	2	/	2	/	2
21	Dual gears	/	1	/	1	/	1	/	1	/	1	/	1	/	1	/	1
22	Reduction gear fixed frame	/	1	/	1	/	1	/	1	/	1	/	1	/	1	/	1
23	Reduction gear fixed stator	/	1	/	1	/	1	/	1	/	1	/	1	/	1	/	1
24	Cross Recessed Countersunk Screw	/	2	/	2	/	2	/	2	/	2	/	2	/	2	/	2

VII-4 CHAIN ASSEMBLY DRAWING

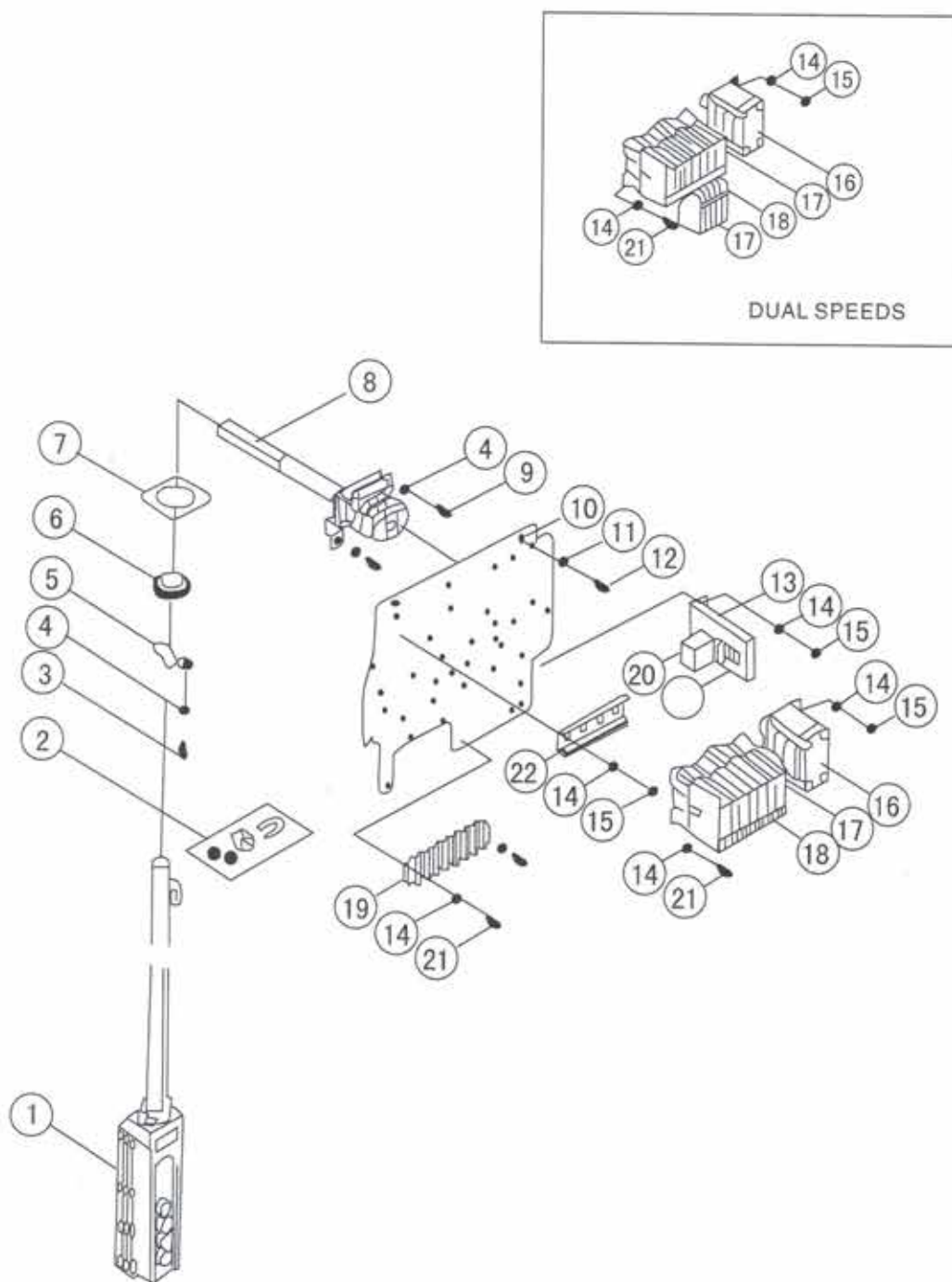


VII-4 CHAIN ASSEMBLY PARTS LIST

S = SINGLE SPEED ; D = DUAL SPEEDS

Item	Spare Parts Name	Every Type Hoist Use Quantity															
		00301,00501		0101		01501		0201		02501		0301		0502		07503	
		S	D	S	D	S	D	S	D	S	D	S	D	S	D	S	D
1	socket head cap screws	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
2	Oriented Iron	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	Slotted countersunk head screws	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
4	Chain Guide	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	Pulley Shaft	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	Oriented Pulley	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	Spring Washer	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
8	socket head cap screws	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
9	split pin	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
10	Limit switches connecting shaft	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
11	Axis of the Middle pin	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
12	Middle pin of guide frame	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
13	Guide frame assembly	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
14	Chain bag	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
15	Limit Spring	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
16	socket head cap screws	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
17	Chain split ring	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
18	Spring Washer	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
19	Chain	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m	3m
20	Nut	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
21	Spring Washer	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2

VII-5 Gear box assembly drawing

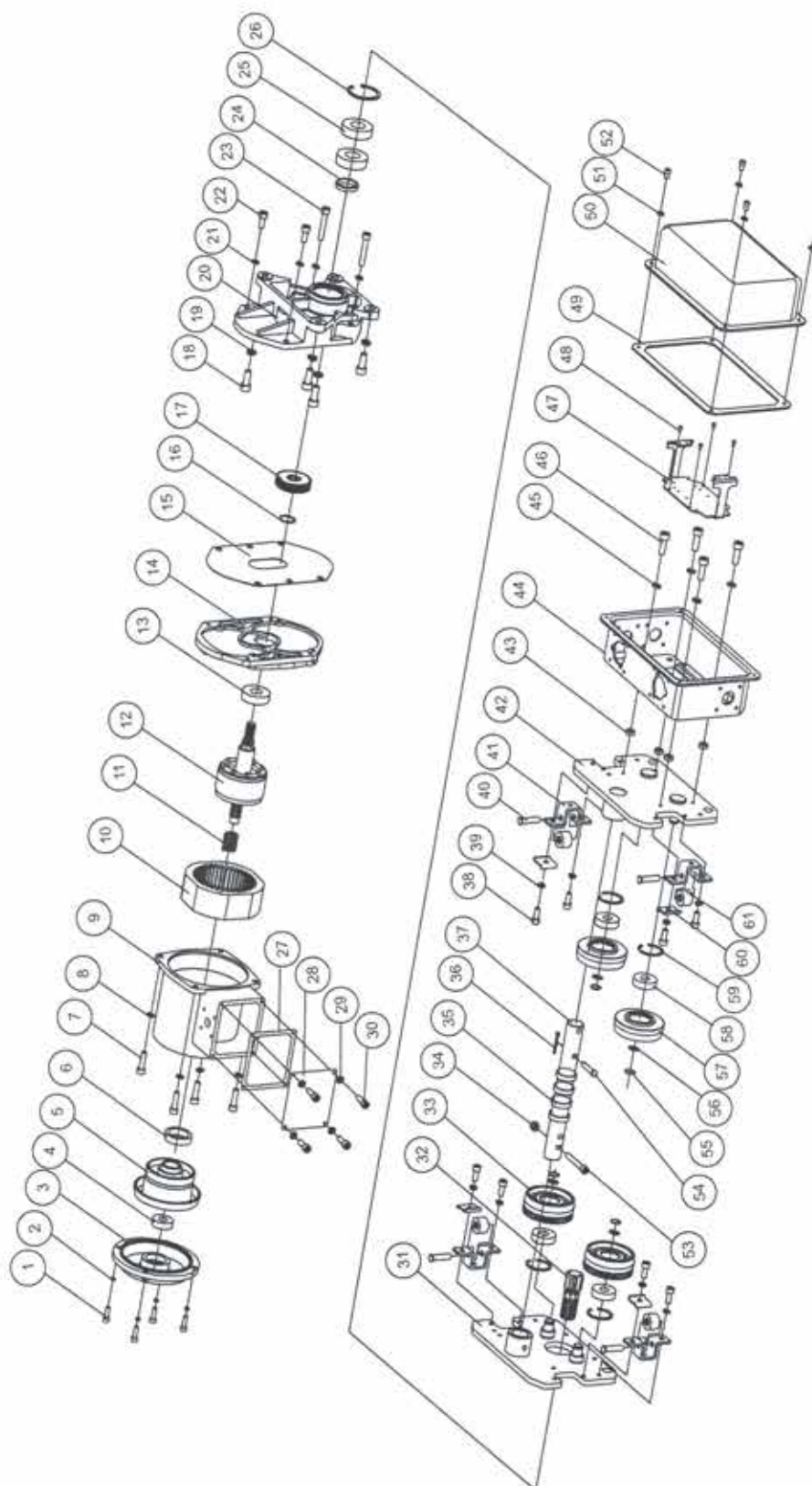


VII-5 GEAR BOX ASSEMBLY PARTS LIST

S = SINGLE SPEED ; D = DUAL SPEEDS

Item	Spare Parts Name	Every Type Hoist Use Quantity															
		00301, 00501		0101		01501		0201		02501		0301		0502		07503	
		S	D	S	D	S	D	S	D	S	D	S	D	S	D	S	D
1	Control switch	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	Tightrope Buckle	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	Round cross screw M6*12	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
4	Spring Washer	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
5	Wiring fixed ring accessories	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	Cable sheath	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	Fixed wiring ring	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
8	Connecting shaft assembly for control switch	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
9	Round cross screw M6*10	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
10	Electrical installation plate	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
11	Spring Washer ?6	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
12	Round cross screw M6*15	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
13	Reverse bracket B	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
14	Spring Washer $\phi 4$	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
15	Round cross screw M4*10	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
16	Transformer	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
17	Electromagnetic contactor	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
18	Mechanical linkage	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
19	Terminal block	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
20	Antiphase protector	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
21	Round cross screw M4*15	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
22	Fixed rail	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

VII-7 ELECTRIC TROLLEY ASSEMBLY DRAWING

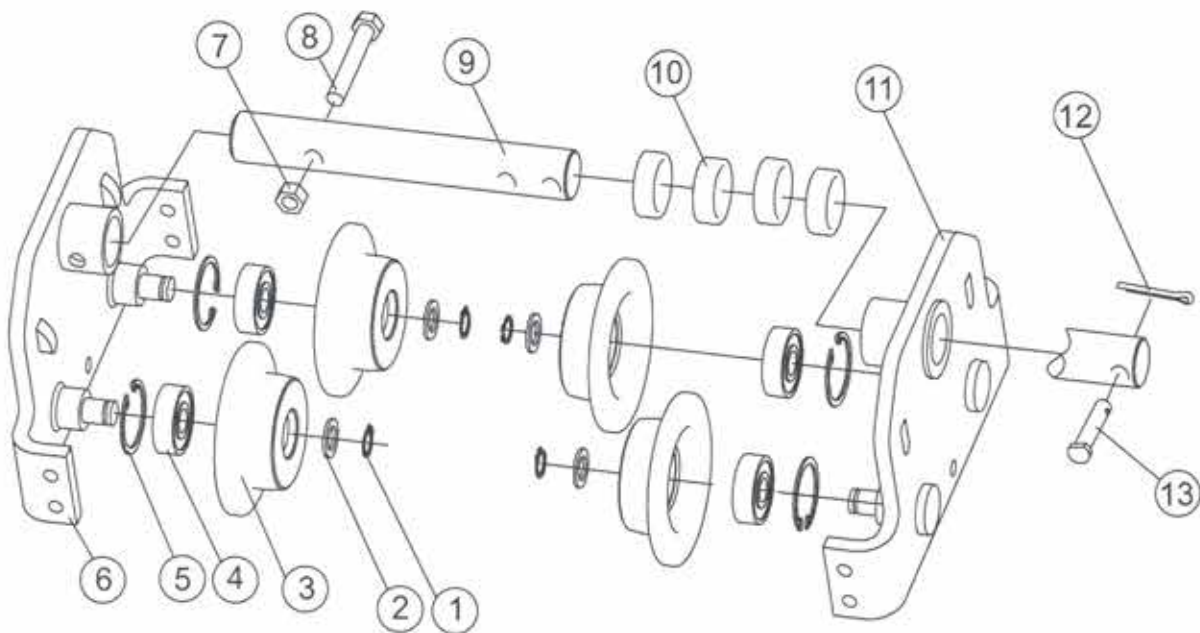


VII-7 ELECTRIC TROLLEY ASSEMBLY PARTS LIST

S = SINGLE SPEED ; D = DUAL SPEEDS

Item	Spare Parts Name	Every Type Hoist Use Quantity						Item	Spare Parts Name	Every Type Hoist Use Quantity					
		0.5t	01t	02t	03t	05t	7.5t			0.5t	01t	02t	03t	05t	7.5t
1	socket head cap screws/M5*20	4	4	4	4	4	4	32	spline gear shaft	1	1	1	1	1	1
2	spring washer $\phi 5$	4	4	4	4	4	4	33	toothed driving wheel	2	2	2	2	2	2
3	motor back end cover	1	1	1	1	1	1	34	self-locking nut	1	1	1	1	1	1
4	deep groove ball bearing/202	1	1	1	1	1	1	35	adjusting spacer	8	8	8	8	8	8
5	5-Brake Assembly	1	1	1	1	1	1	36	split pin	1	1	1	1	1	1
6	rubber circlip	1	1	1	1	1	1	37	trolley shaft	1	1	1	1	1	1
7	socket head cap screws	4	4	4	4	4	4	38	socket head cap screw	8	8	8	8	8	8
8	spring washer	4	4	4	4	4	4	39	spring washer	8	8	8	8	8	8
9	motor shell	1	1	1	1	1	1	40	auxiliary wheel shaft	4	4	4	4	4	4
10	motor stator	1	1	1	1	1	1	41	auxiliary wheel bracket	4	4	4	4	4	4
11	brake spring	1	1	1	1	1	1	42	trolley subplate assembly	1	1	1	1	1	1
12	rotor assembly	1	1	1	1	1	1	43	gasket	4	4	4	4	4	4
13	deep groove ball bearing	1	1	1	1	1	1	44	electric control case shell assembly	1	1	1	1	1	1
14	motor bottom plate	1	1	1	1	1	1	45	spring washer	4	4	4	4	4	4
15	motor bottom plate gasket	1	1	1	1	1	1	46	socket head cap screw	4	4	4	4	4	4
16	circlip for shaft	1	1	1	1	1	1	47	wire connection bracket	1	1	1	1	1	1
17	gear	1	1	1	1	1	1	48	Cross Recess Head Screw	4	4	4	4	4	4
18	socket head cap screw	4	4	4	4	4	4	49	electric control case shell gasket	1	1	1	1	1	1
19	spring washer	4	4	4	4	4	4	50	electric control case shell	1	1	1	1	1	1
20	gear box	1	1	1	1	1	1	51	spring washer	4	4	4	4	4	4
21	spring washer	4	4	4	4	4	4	52	socket head cap screw	4	4	4	4	4	4
22	socket head cap screws	2	2	2	2	2	2	53	socket head cap screw	1	1	1	1	1	1
23	socket head cap screws	2	2	2	2	2	2	54	trolley shaft dowel pin	1	1	1	1	1	1
24	circlip	1	1	1	1	1	1	55	circlip for shaft	4	4	4	4	4	4
25	deep groove ball bearing	2	2	2	2	2	2	56	trolley wheel washer	4	4	4	4	4	4
26	circlip for hole	1	1	1	1	1	1	57	trolley wheel	2	2	2	2	2	2
27	side plate gasket	1	1	1	1	1	1	58	Deep groove ball rolling bearing	4	4	4	4	4	4
28	side plate	1	1	1	1	1	1	59	circlip for hole	4	4	4	4	4	4
29	spring washer	4	4	4	4	4	4	60	auxiliary wheel bracket gasket	4	4	4	4	4	4
30	socket head cap screws	4	4	4	4	4	4	61	trolley auxiliary wheel	4	4	4	4	4	4
31	trolley main plate assembly	1	1	1	1	1	1								

VII-8 TROLLEY ASSEMBLY DRAWING



VII-8 TROLLEY ASSEMBLY PARTS LIST

Item	Spare Parts Name	Every Type Hoist Use Quantity					
		0.5t	01t	02t	03t	05t	7.5t
1	Circlip for shaft	4	4	4	4	4	4
2	Trolley wheel washer	4	4	4	4	4	4
3	Trolley wheel	4	4	4	4	4	4
4	Deep groove ball bearing	4	4	4	4	4	4
5	Circlip for hole	4	4	4	4	4	4
6	Trolley main plate assembly	1	1	1	1	1	1
7	Self-locking nut	1	1	1	1	1	1
8	Outer hexagonal screw	1	1	1	1	1	1
9	Trolley shaft	1	1	1	1	1	1
10	Adjusting spacer	8	8	8	8	8	8
11	Trolley subplate assembly	1	1	1	1	1	1
12	Split pin	1	1	1	1	1	1
13	Trolley shaft dowel pin	1	1	1	1	1	1